

# BFS Traversal

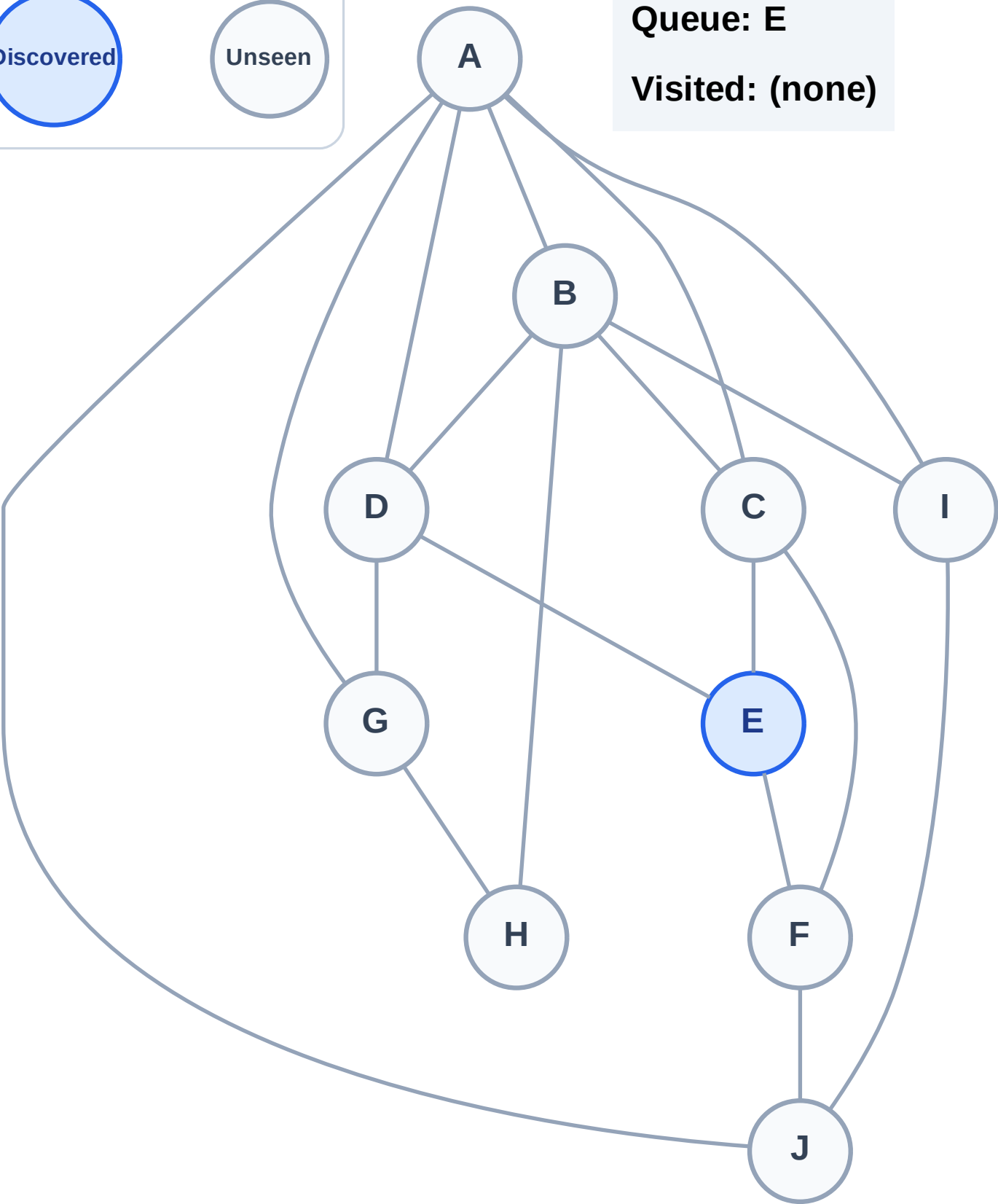
Step 1: Start at E

## Legend



Queue: E

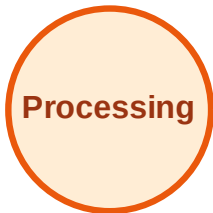
Visited: (none)



# BFS Traversal

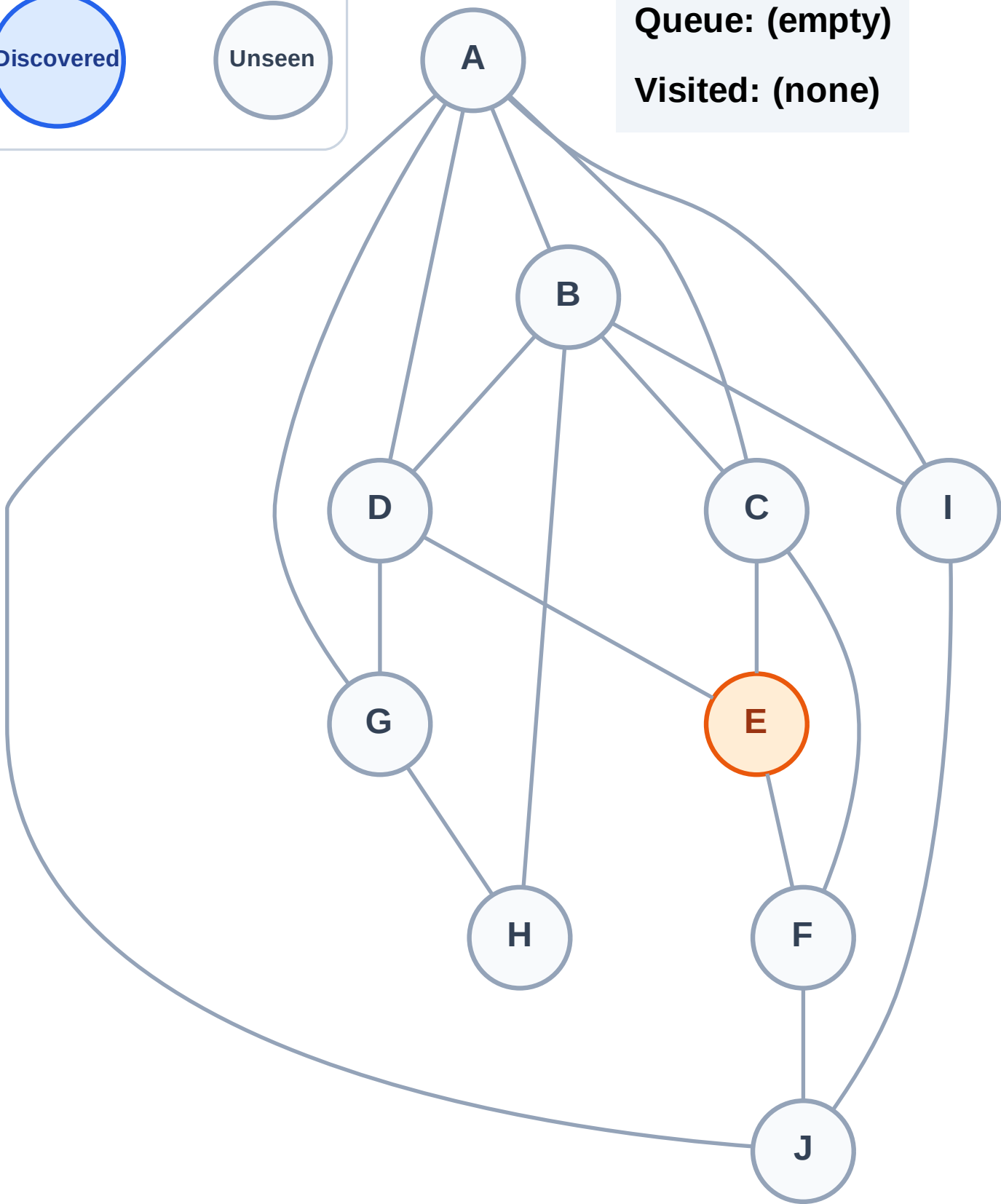
Step 2: Process node E

## Legend



Queue: (empty)

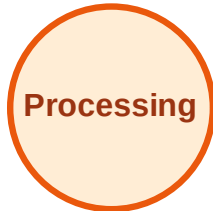
Visited: (none)



# BFS Traversal

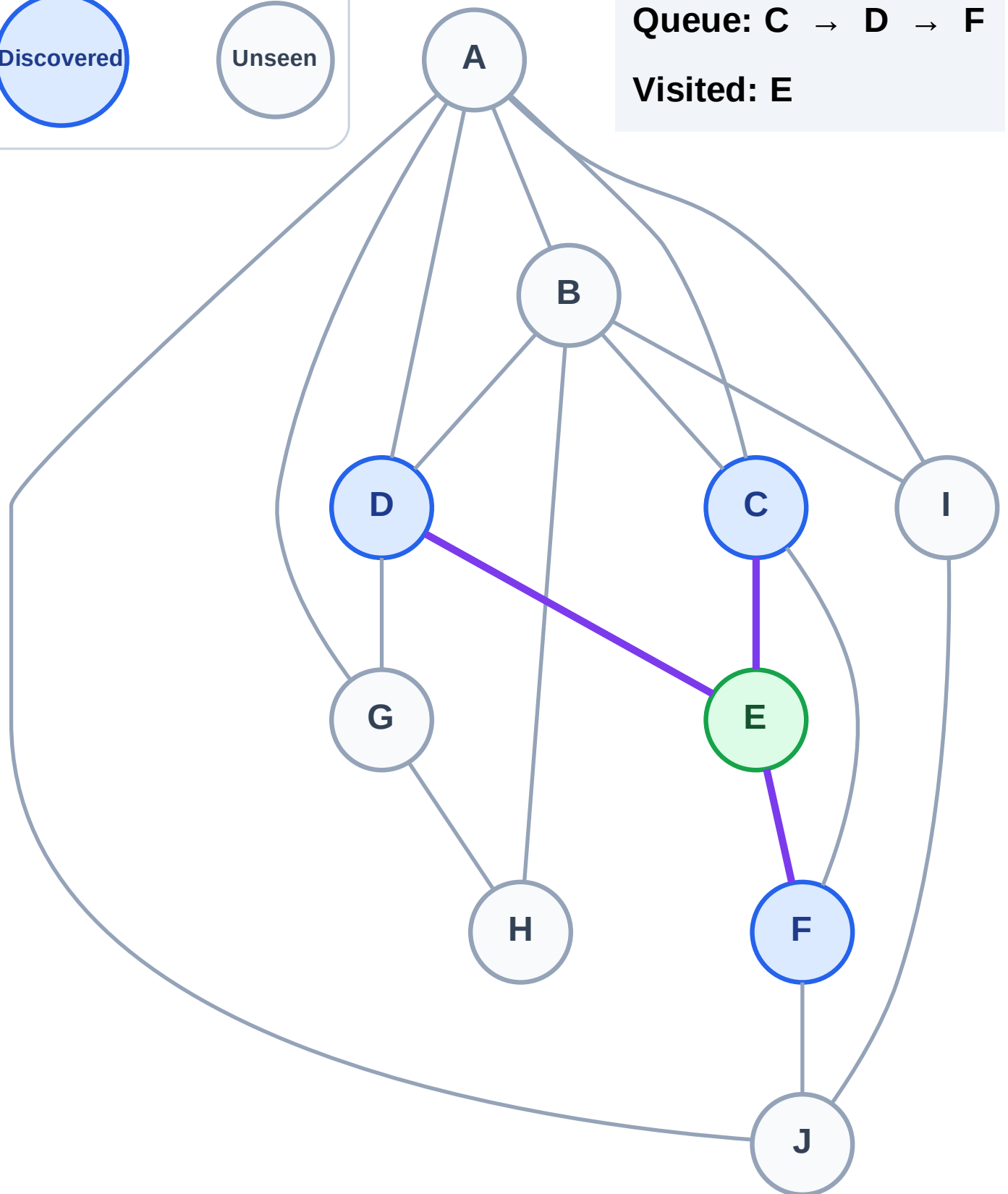
Step 3: Discover C, D, F from E

## Legend



Queue: C → D → F

Visited: E



# BFS Traversal

Step 4: Process node C

Legend

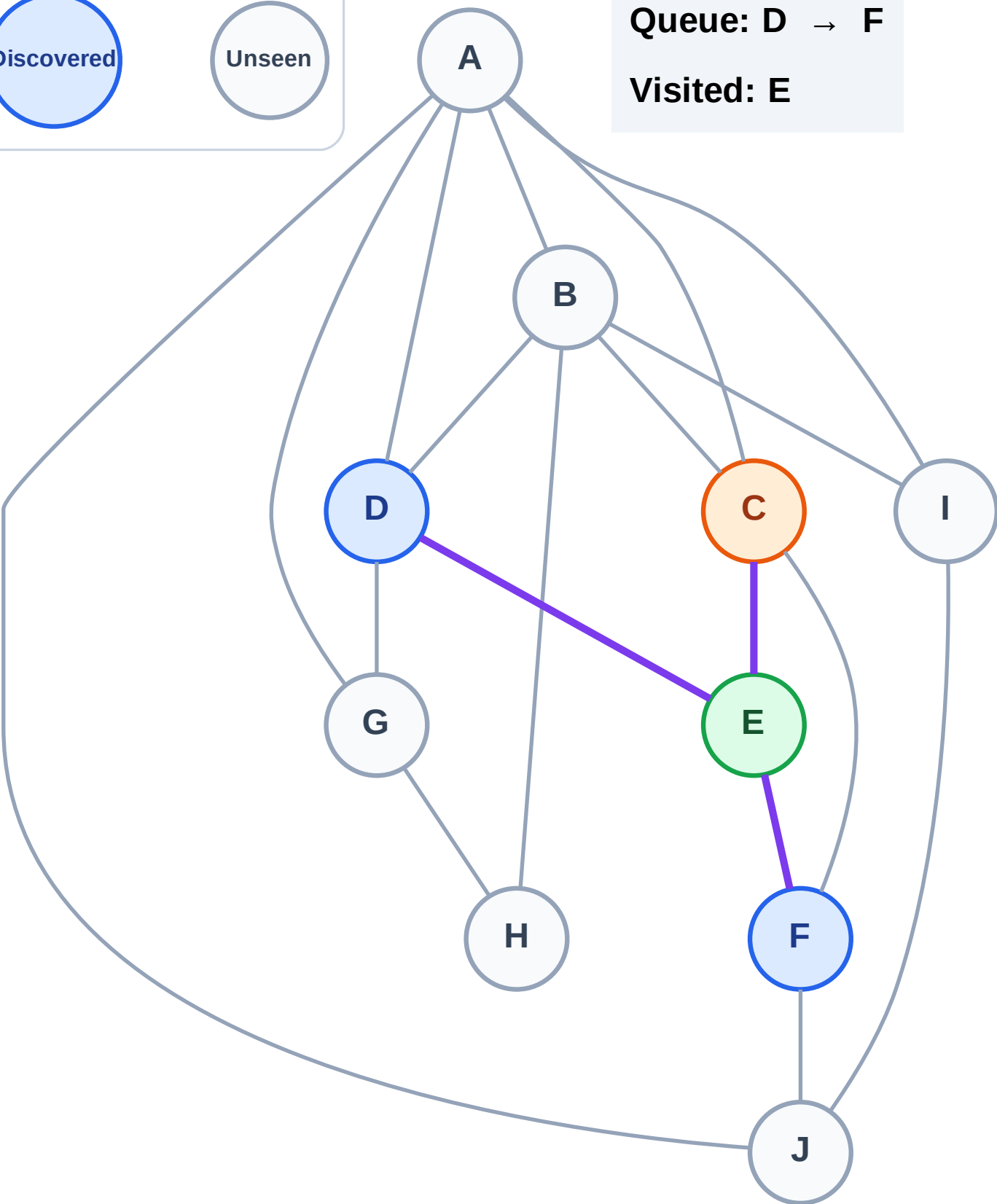
Complete

Processing

Discovered

Unseen

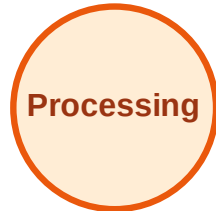
Queue: D → F  
Visited: E



# BFS Traversal

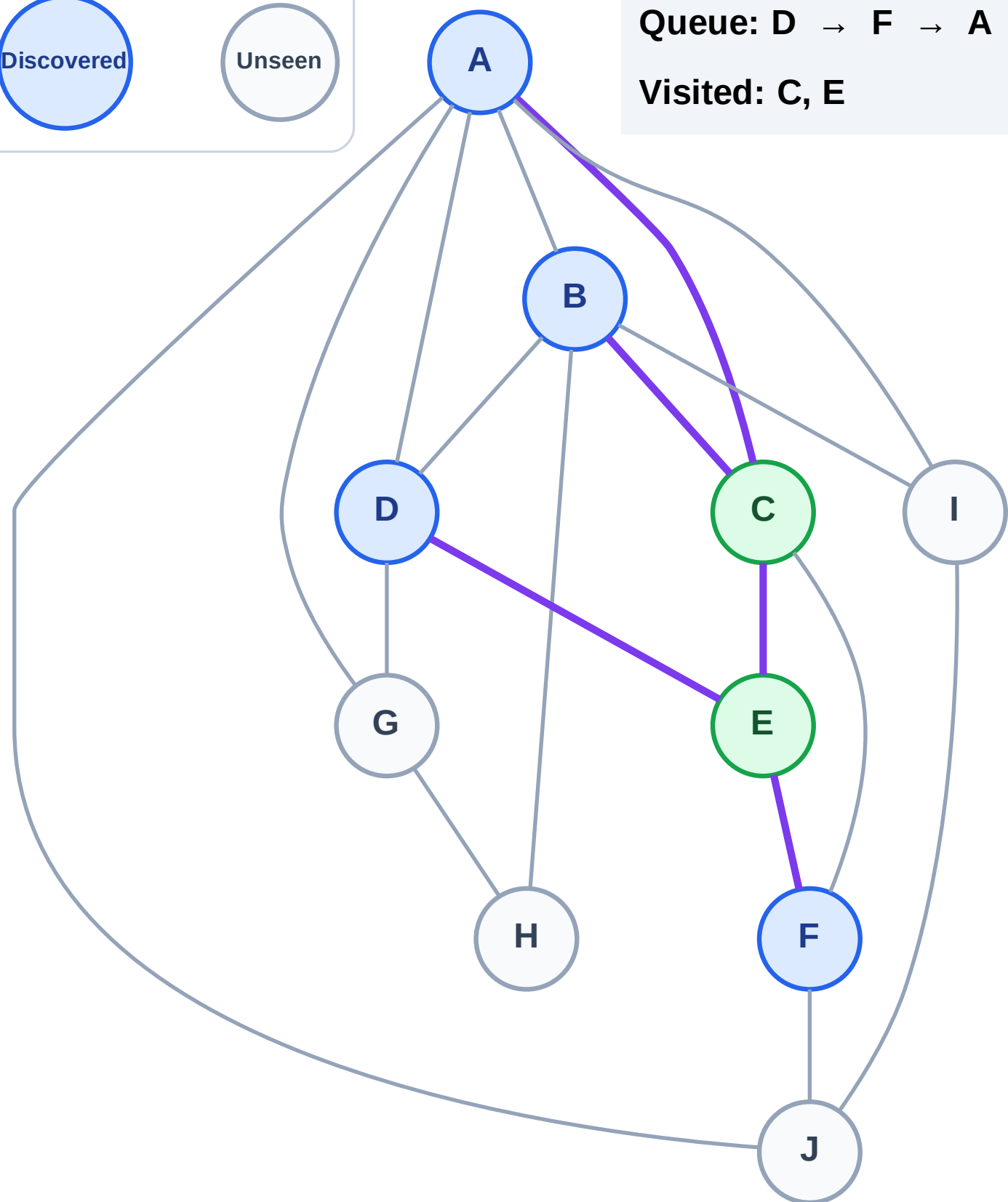
Step 5: Discover A, B from C

## Legend



Queue: D → F → A → B

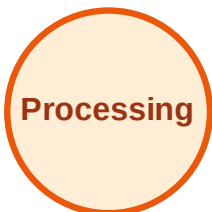
Visited: C, E



# BFS Traversal

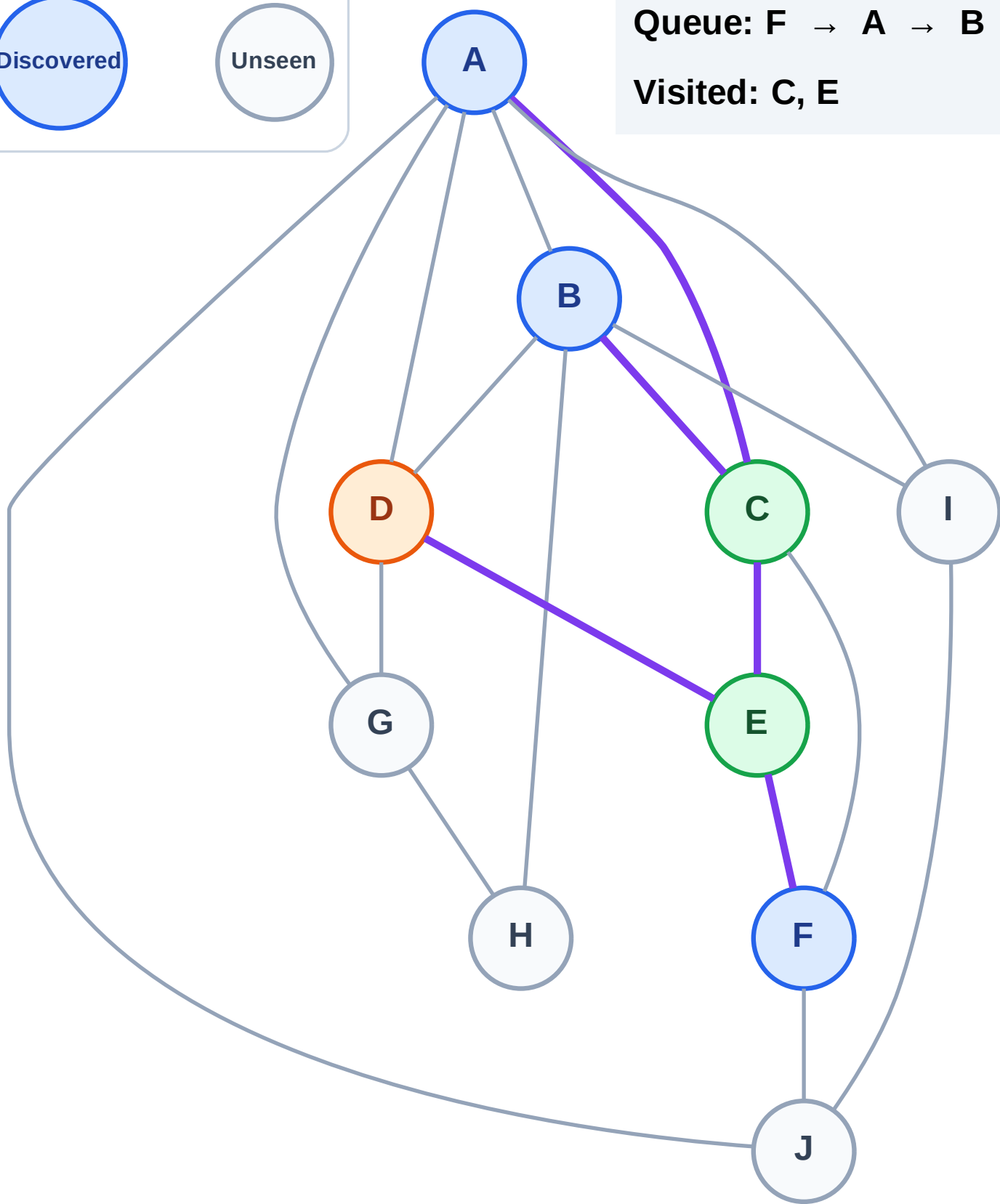
Step 6: Process node D

## Legend



Queue: F → A → B

Visited: C, E



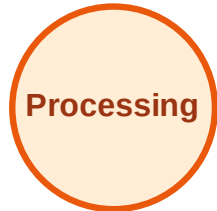
# BFS Traversal

Step 7: Discover G from D

## Legend



Complete



Processing



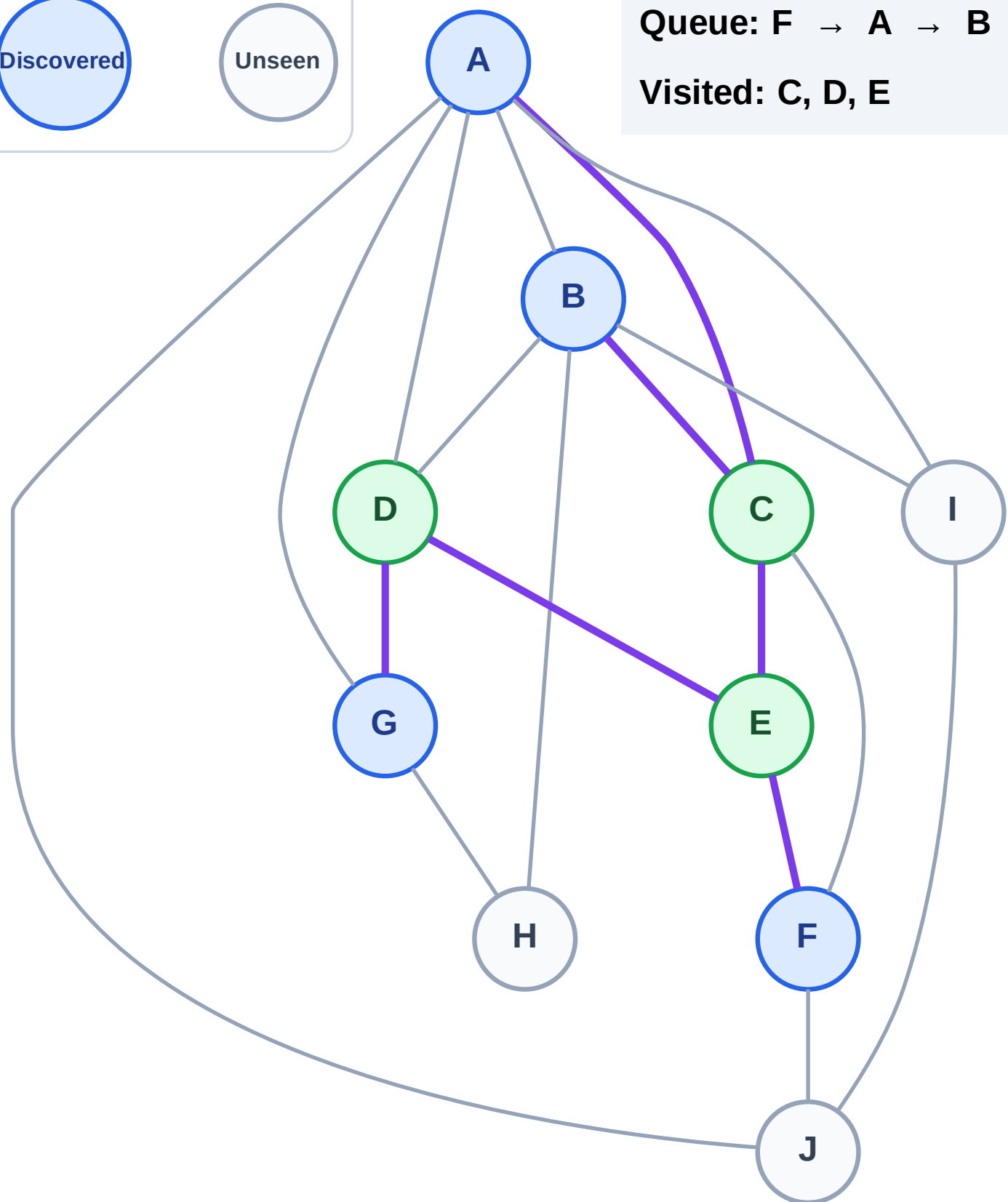
Discovered



Unseen

Queue: F → A → B → G

Visited: C, D, E



# BFS Traversal

Step 8: Process node F

Legend

Complete

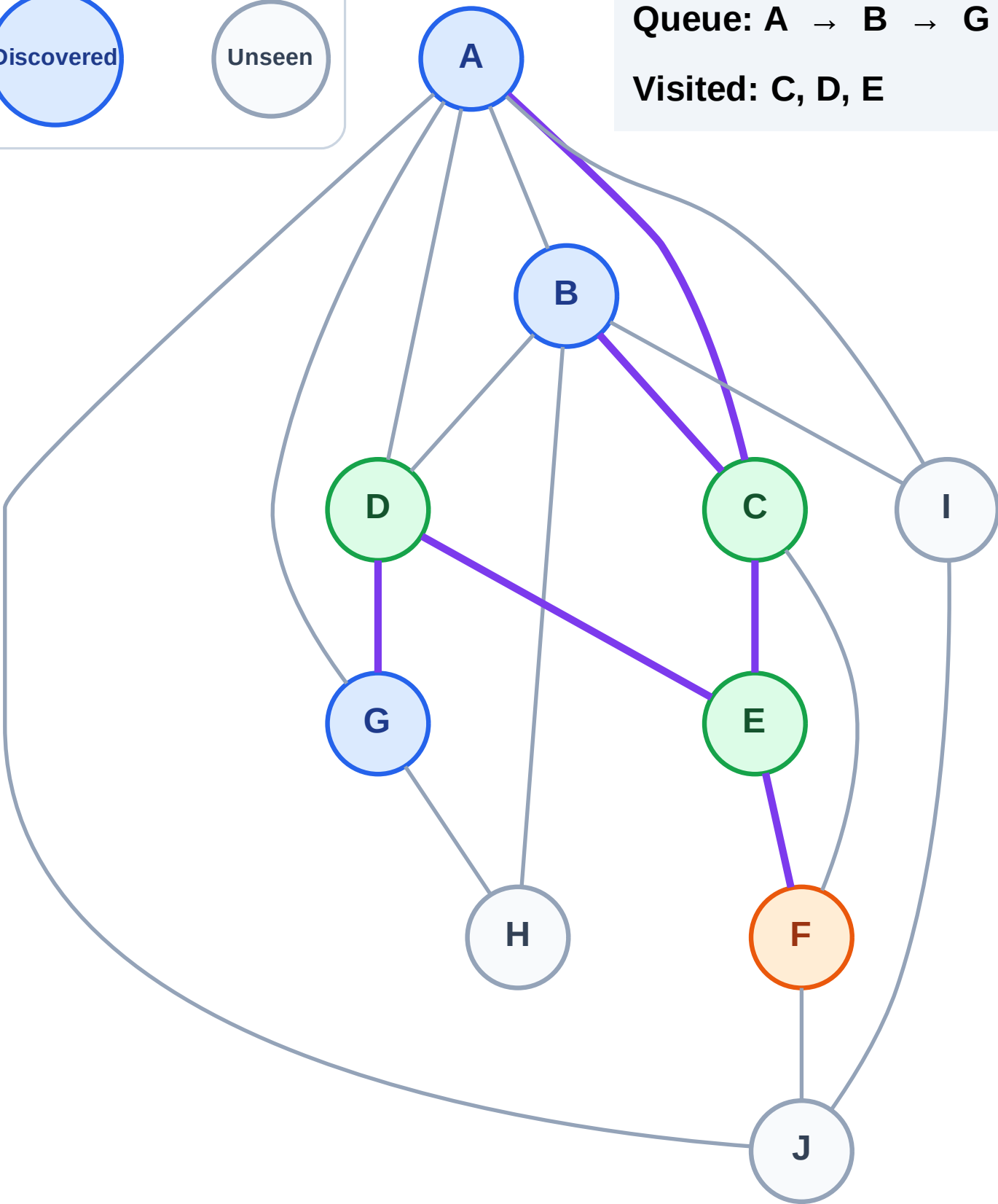
Processing

Discovered

Unseen

Queue: A → B → G

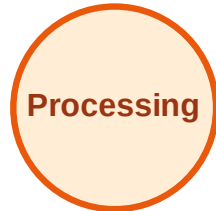
Visited: C, D, E



# BFS Traversal

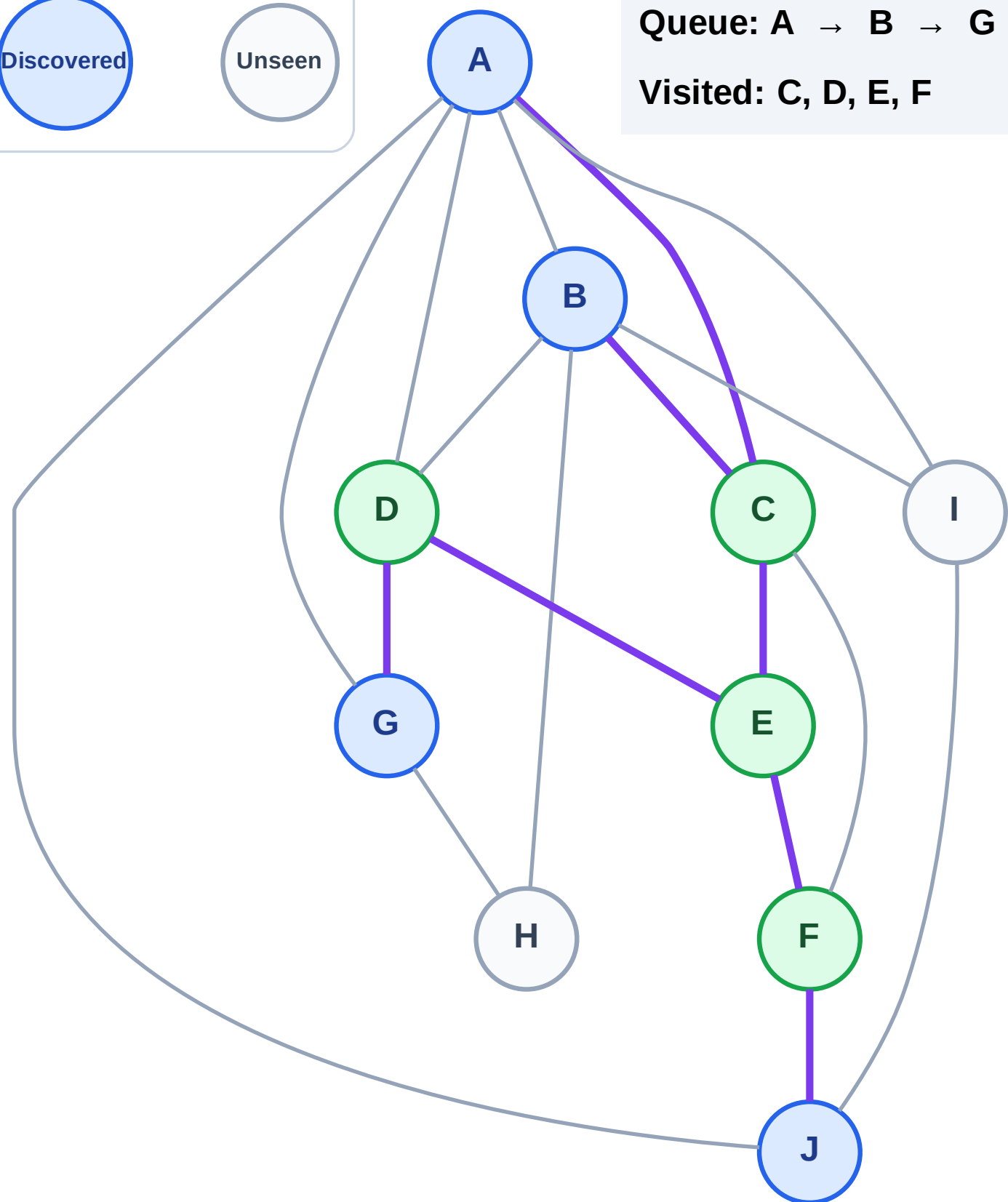
Step 9: Discover J from F

## Legend



Queue: A → B → G → J

Visited: C, D, E, F



# BFS Traversal

Step 10: Process node A

Legend

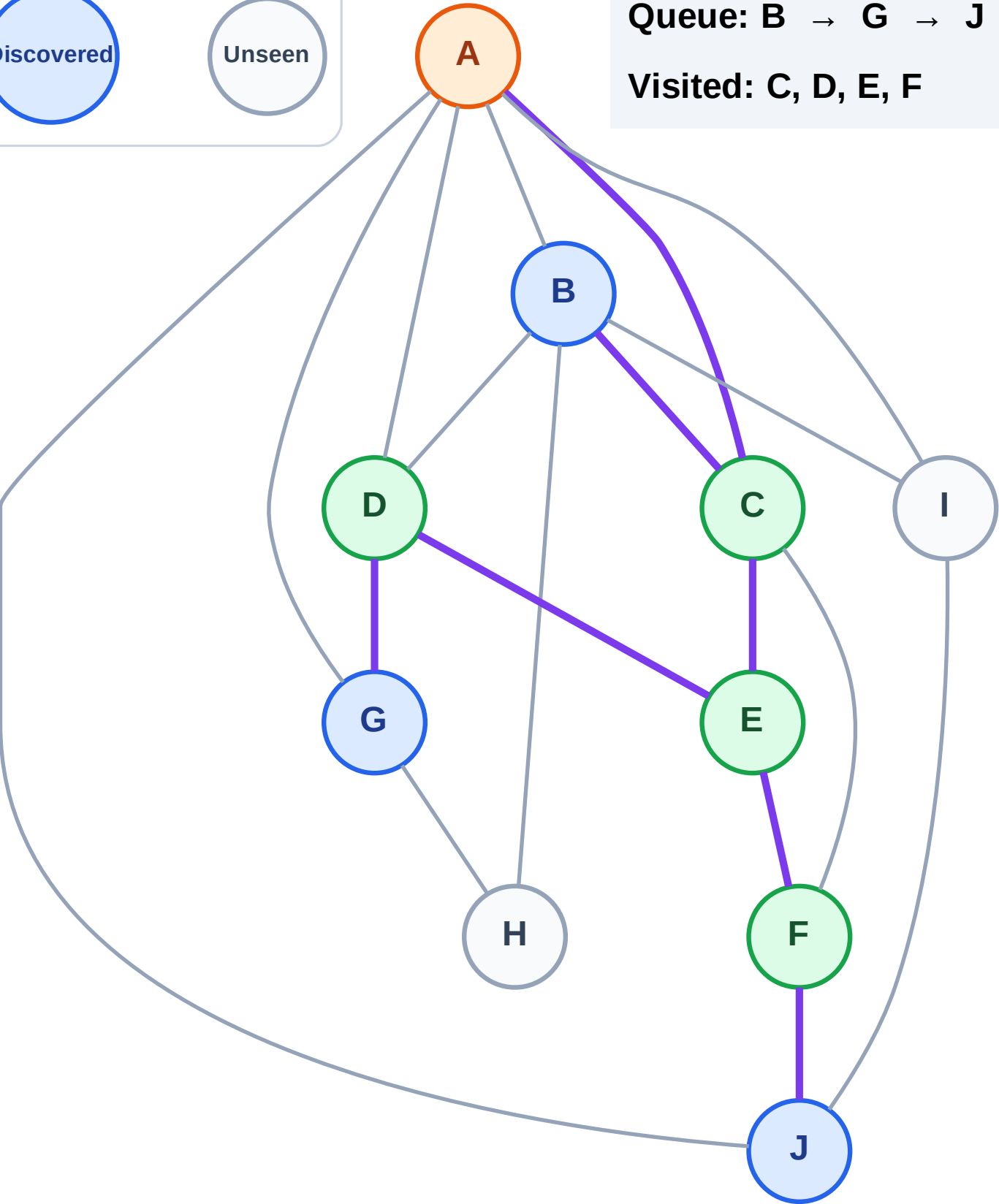
Complete

Processing

Discovered

Unseen

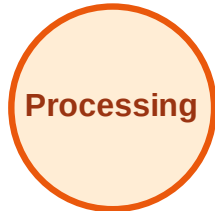
Queue: B → G → J  
Visited: C, D, E, F



# BFS Traversal

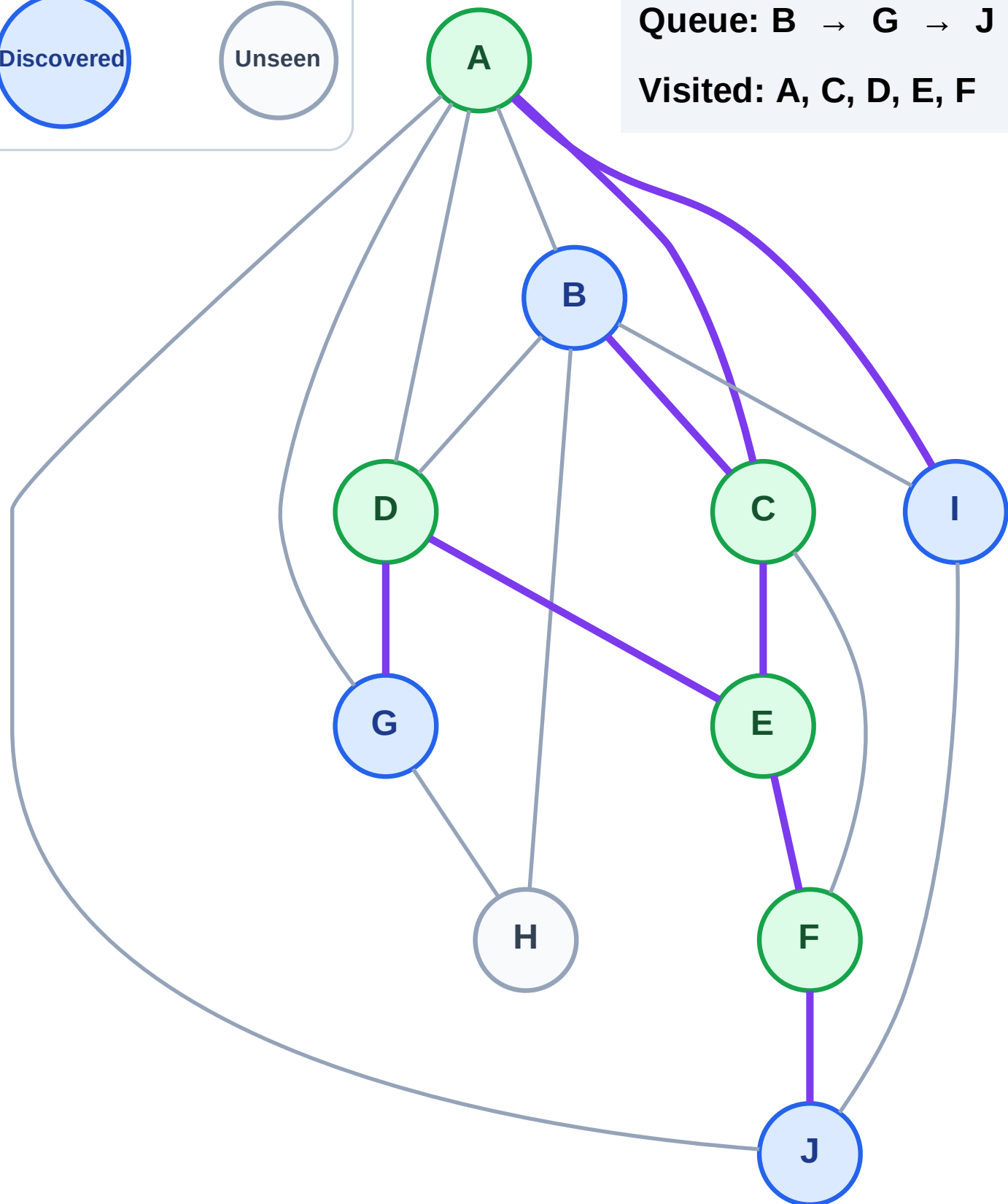
Step 11: Discover I from A

## Legend



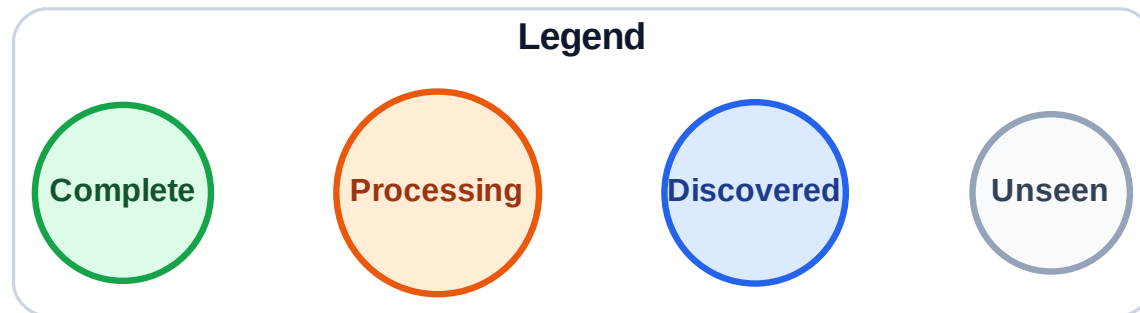
Queue: B → G → J → I

Visited: A, C, D, E, F

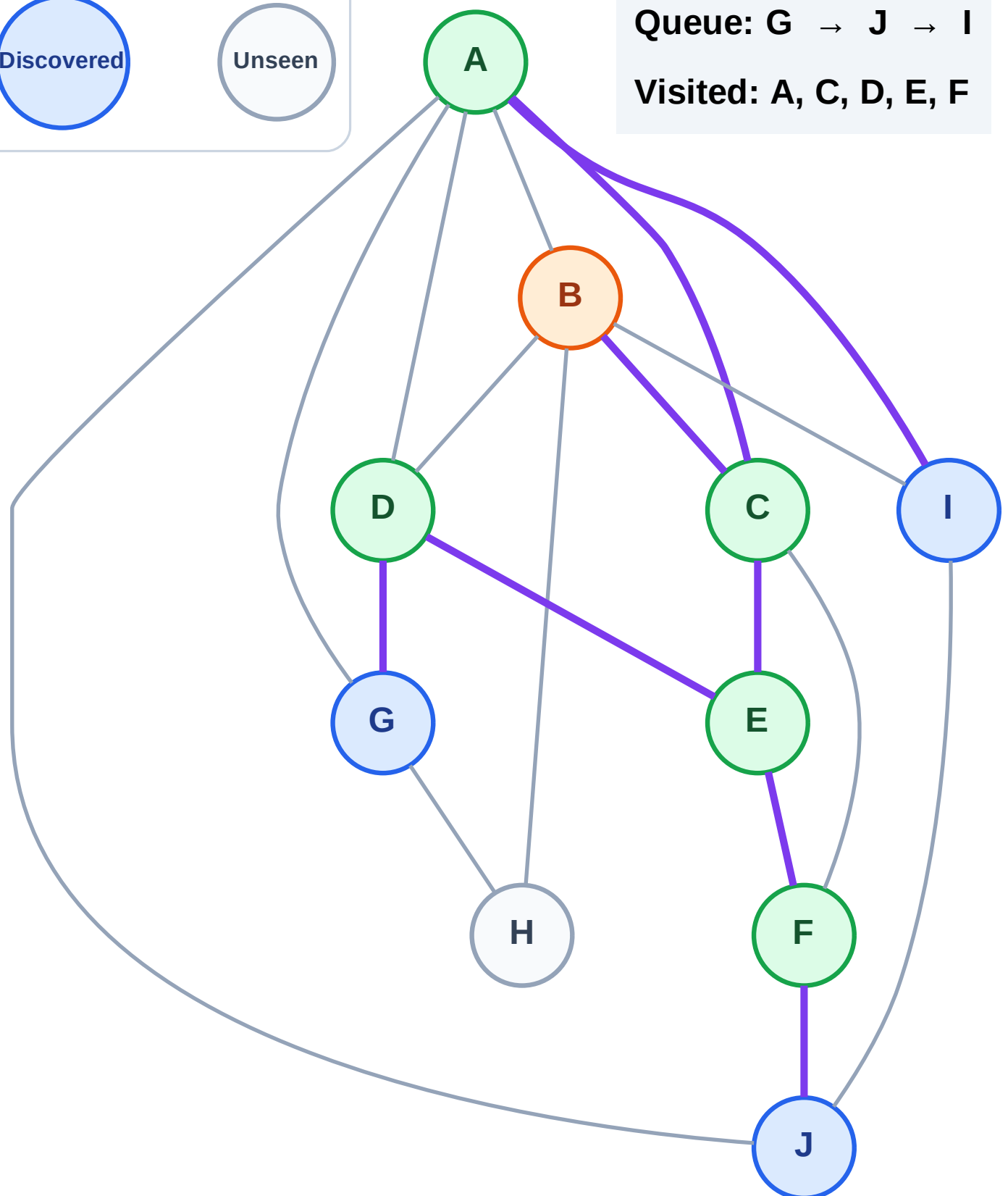


# BFS Traversal

Step 12: Process node B



Queue: G → J → I  
Visited: A, C, D, E, F



# BFS Traversal

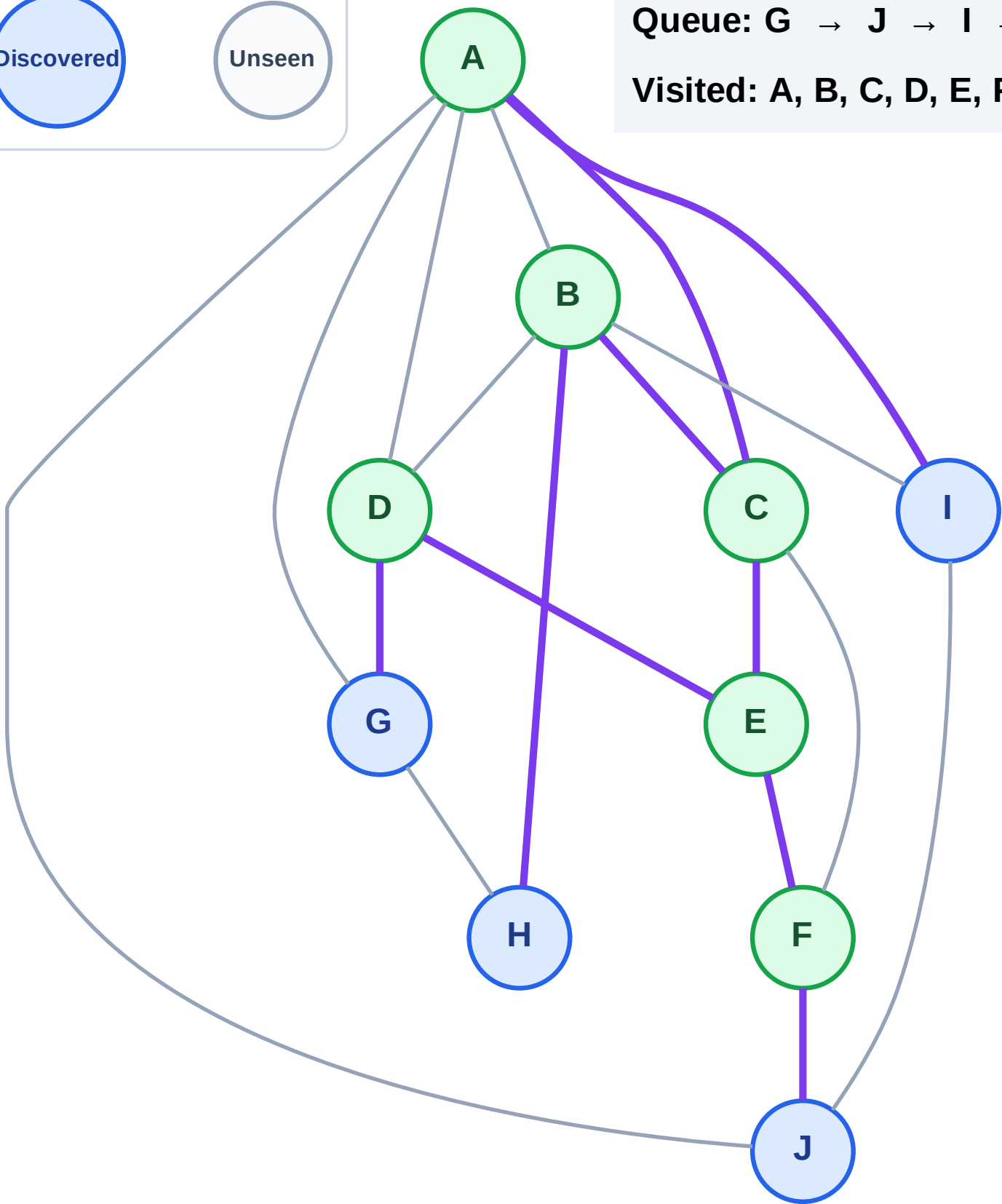
Step 13: Discover H from B

## Legend



Queue: G → J → I → H

Visited: A, B, C, D, E, F



# BFS Traversal

Step 14: Process node G

Legend

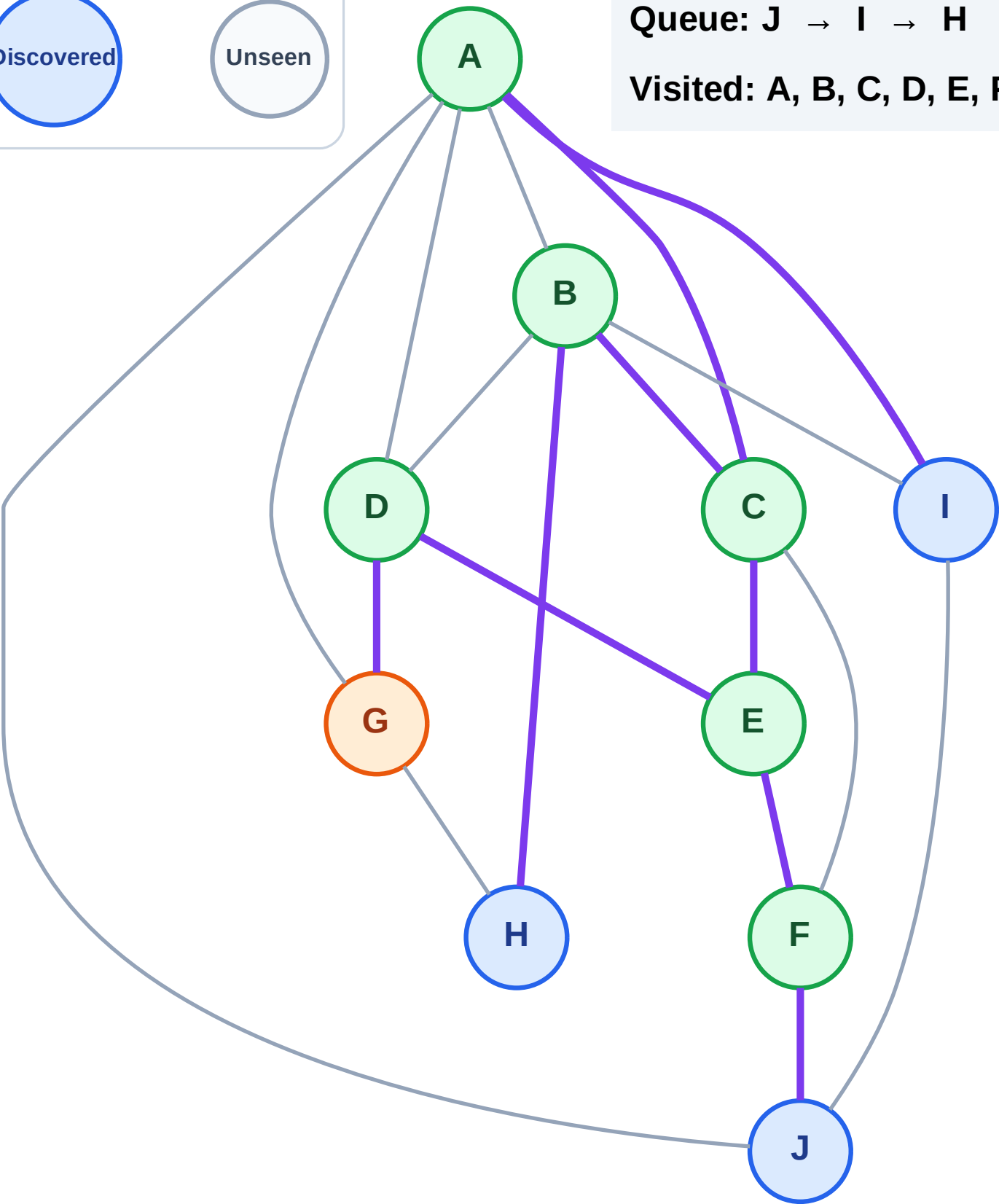
Complete

Processing

Discovered

Unseen

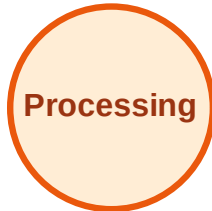
Queue: J → I → H  
Visited: A, B, C, D, E, F



# BFS Traversal

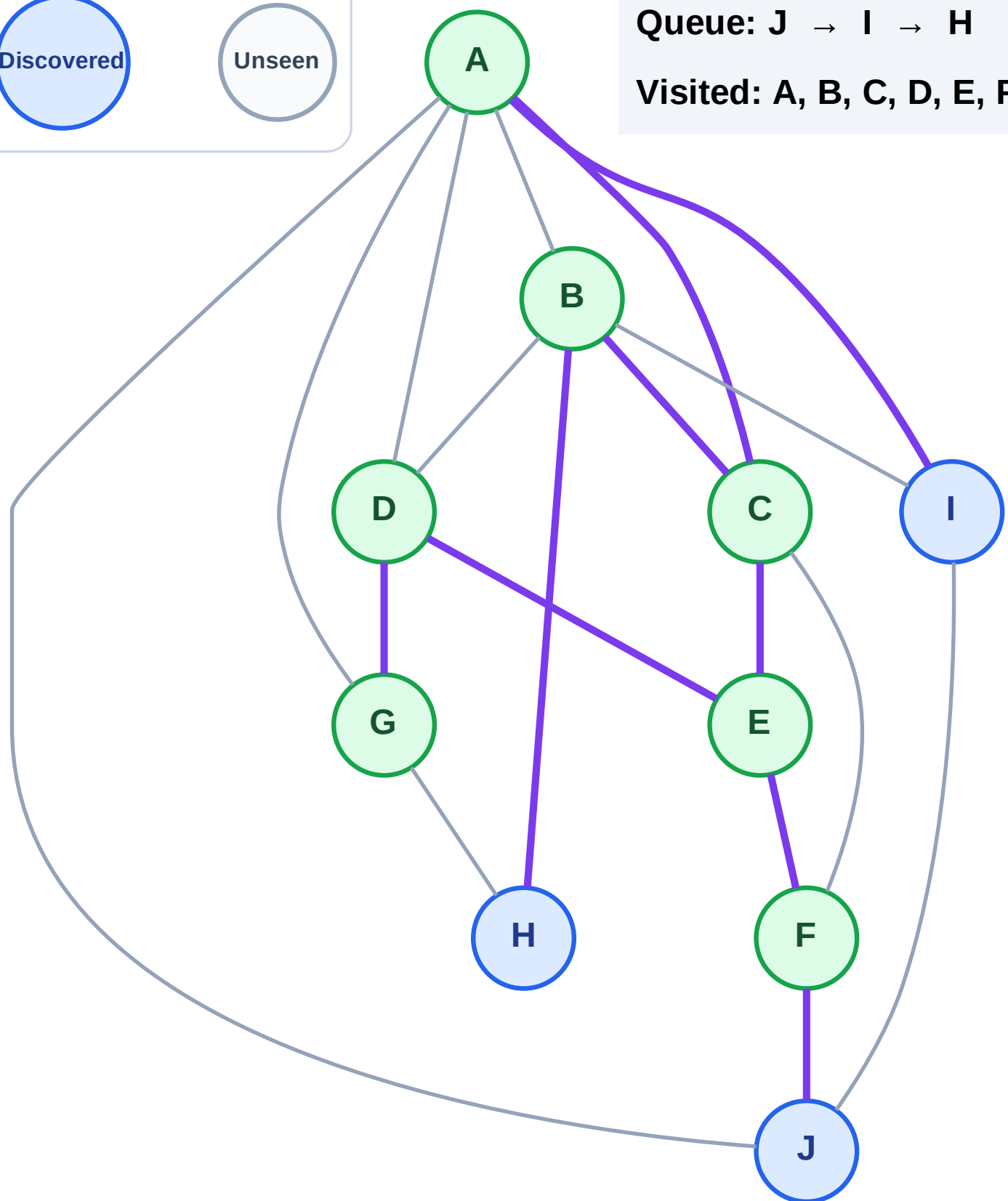
Step 15: No new neighbors from G

## Legend



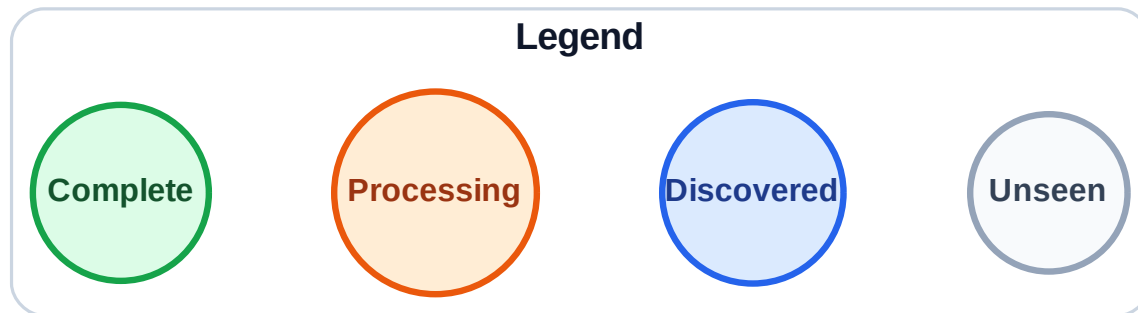
Queue: J → I → H

Visited: A, B, C, D, E, F, G



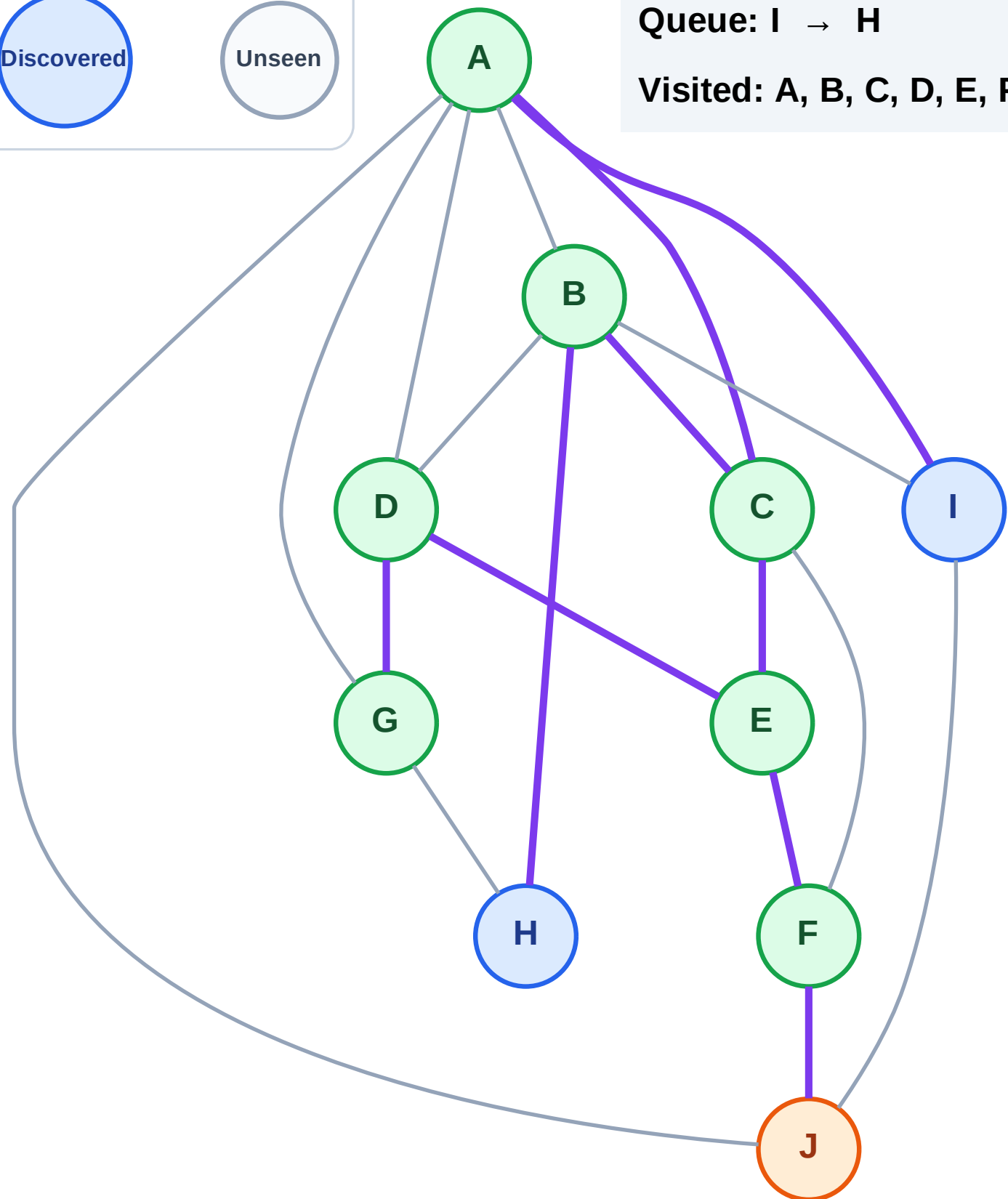
### Step 16: Process node J

### Step 16: Process node J



**Queue: I → H**

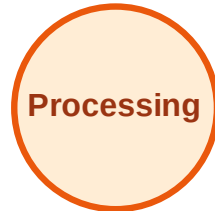
**Visited: A, B, C, D, E, F, G**



# BFS Traversal

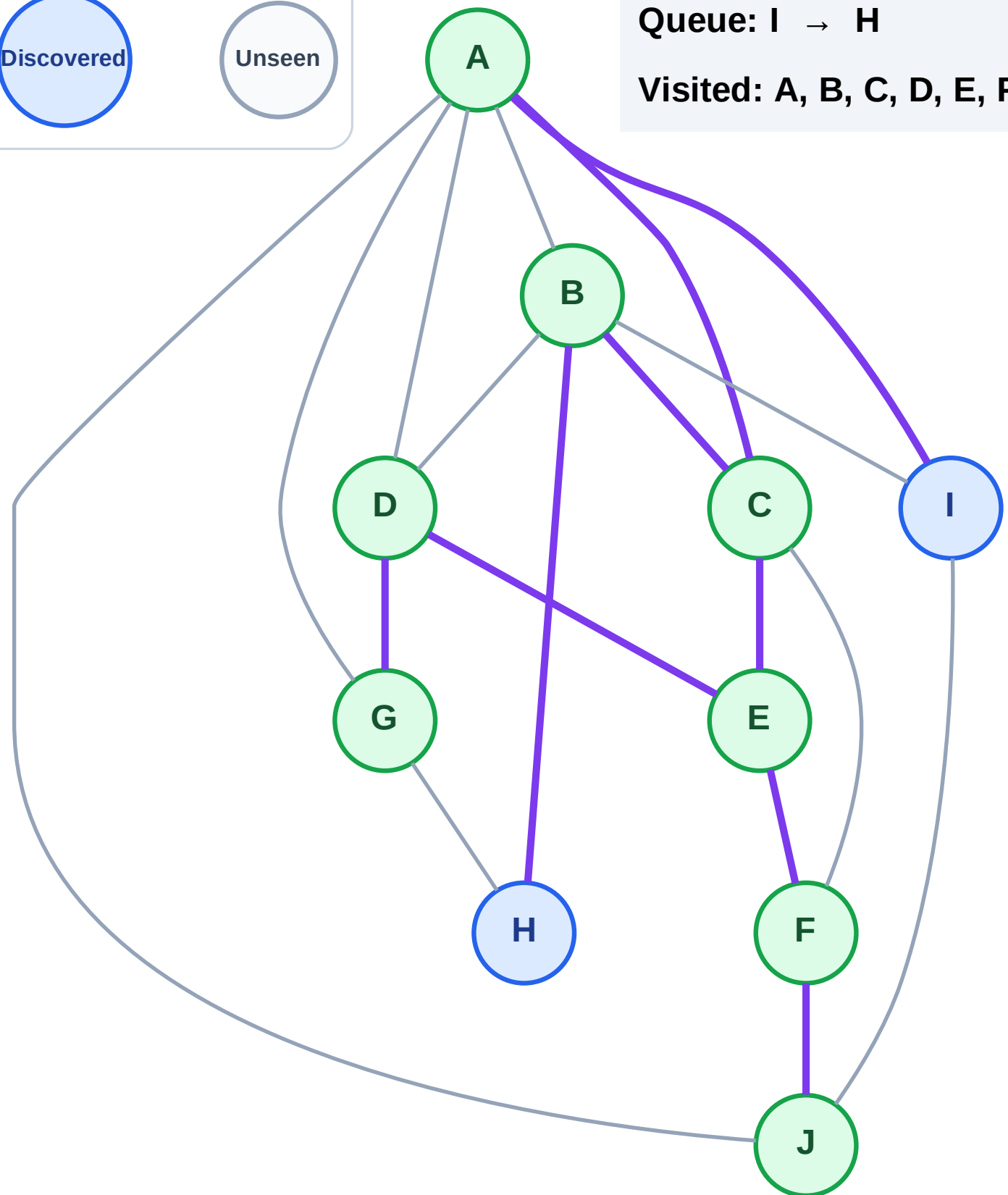
Step 17: No new neighbors from J

## Legend



Queue: I → H

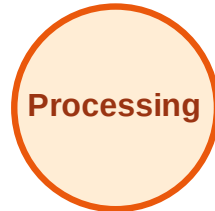
Visited: A, B, C, D, E, F, G, J



# BFS Traversal

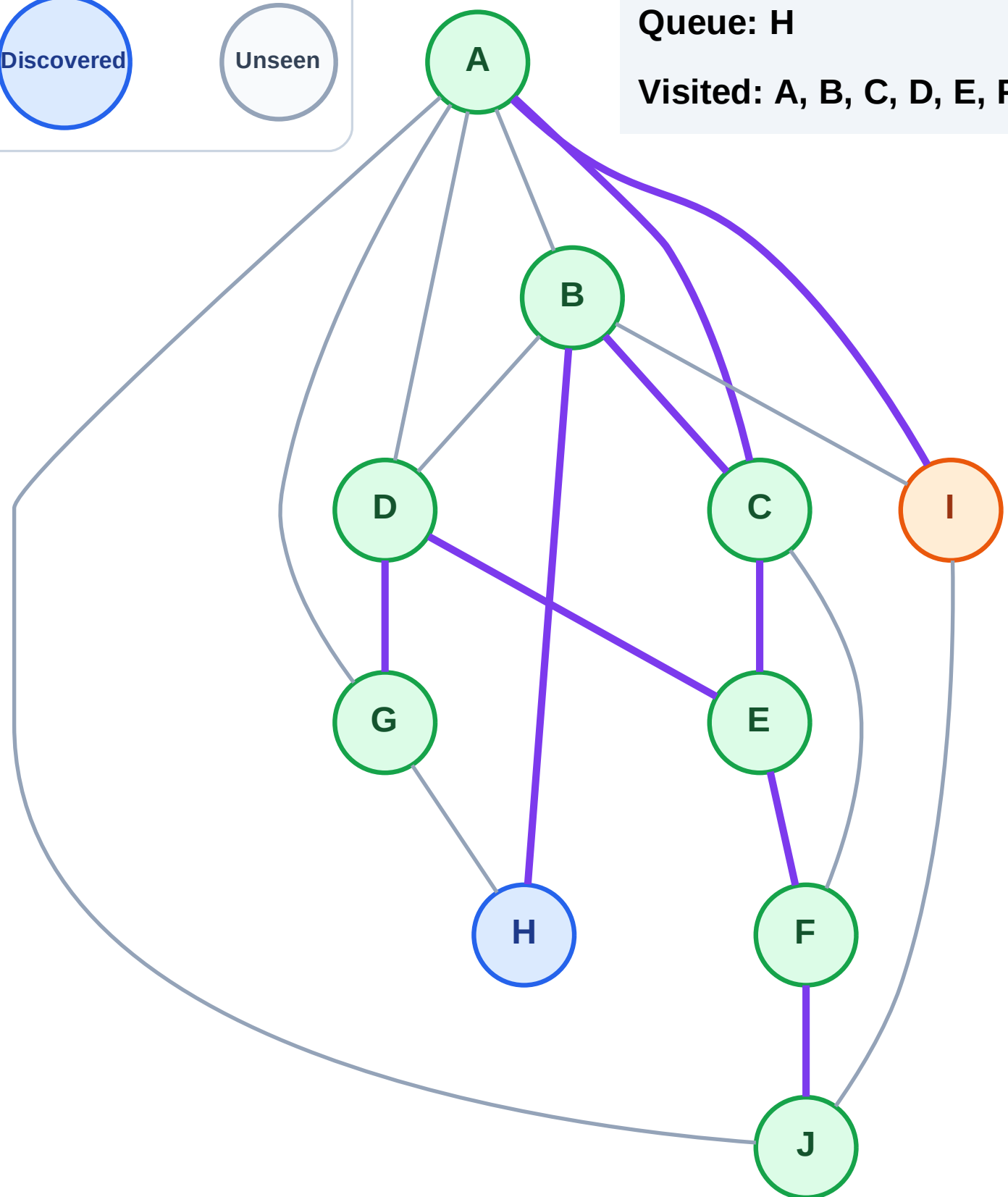
Step 18: Process node I

## Legend



Queue: H

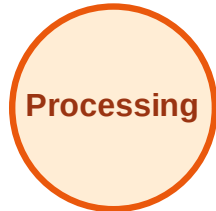
Visited: A, B, C, D, E, F, G, J



# BFS Traversal

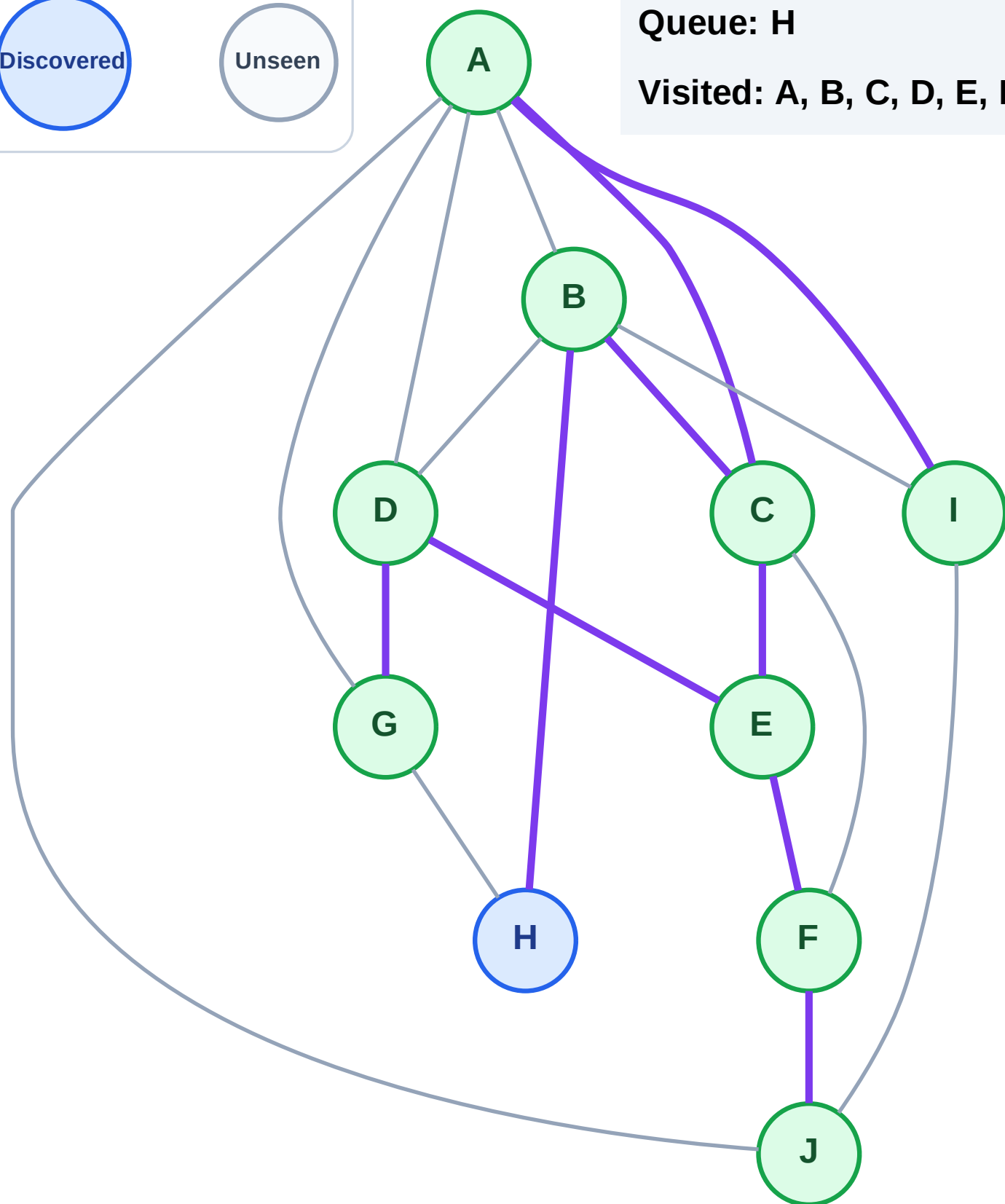
Step 19: No new neighbors from I

## Legend



Queue: H

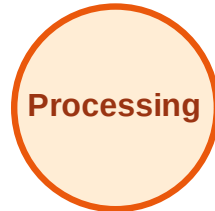
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

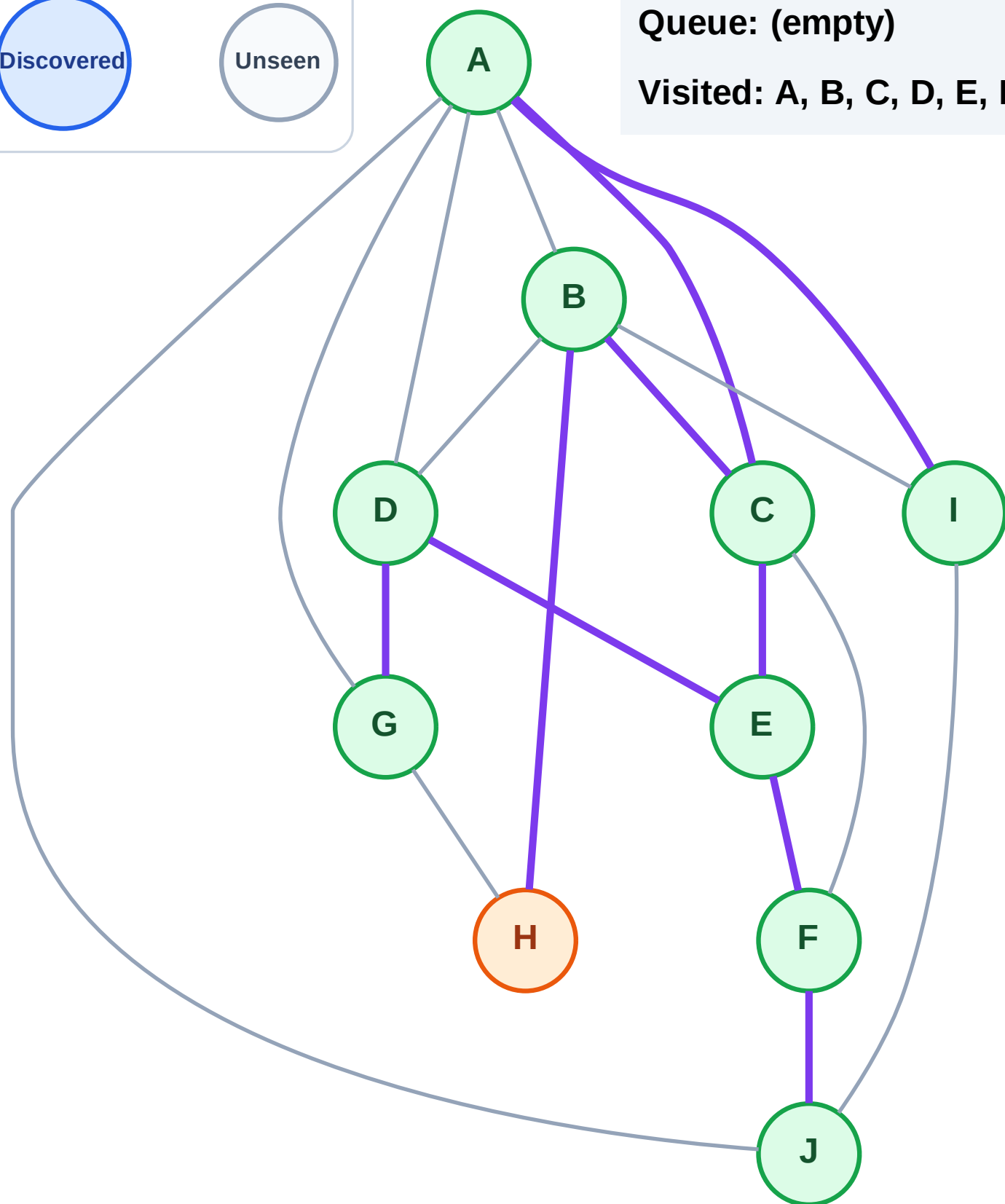
Step 20: Process node H

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

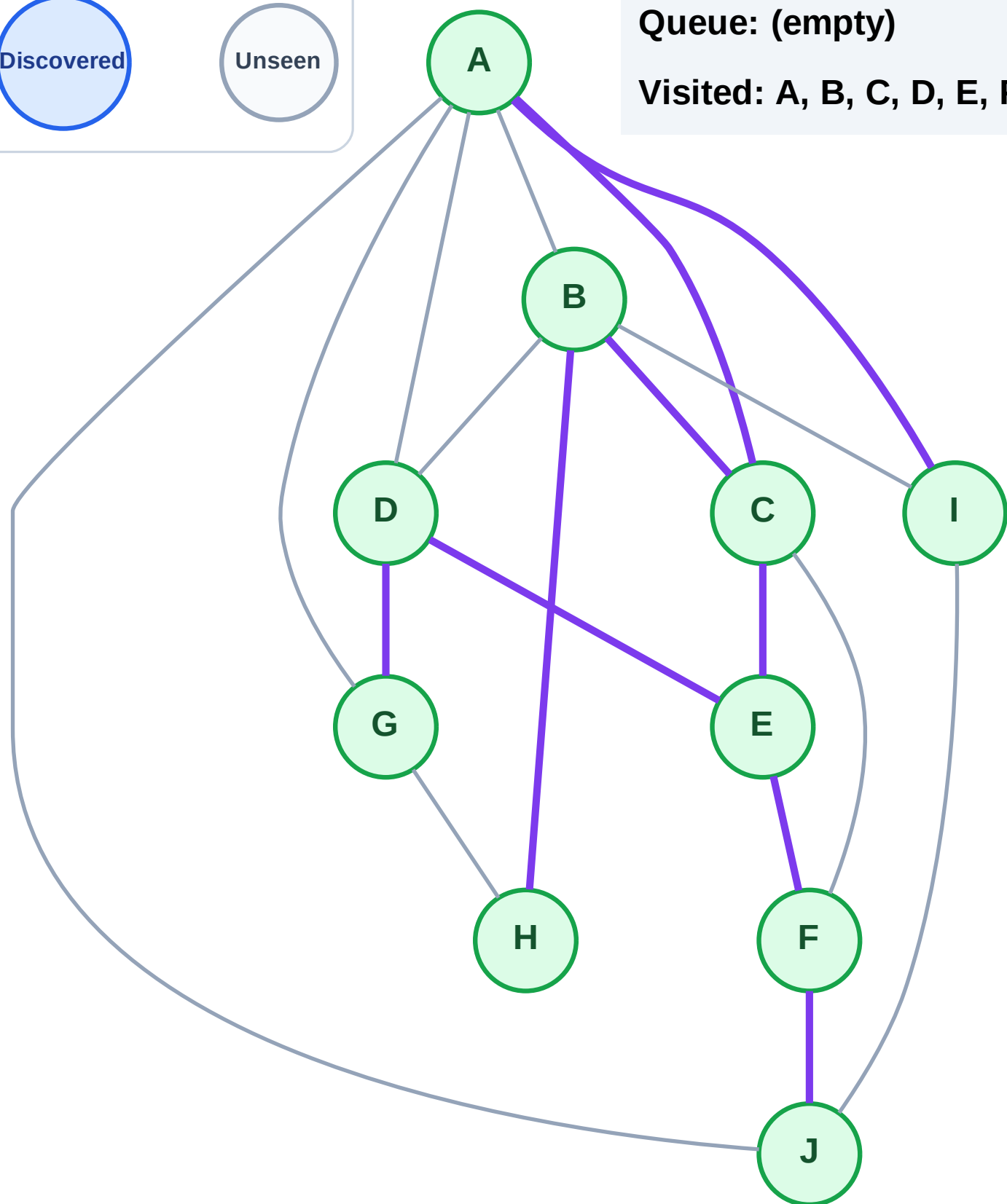
Step 21: No new neighbors from H

## Legend



Queue: (empty)

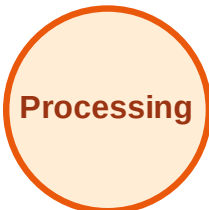
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

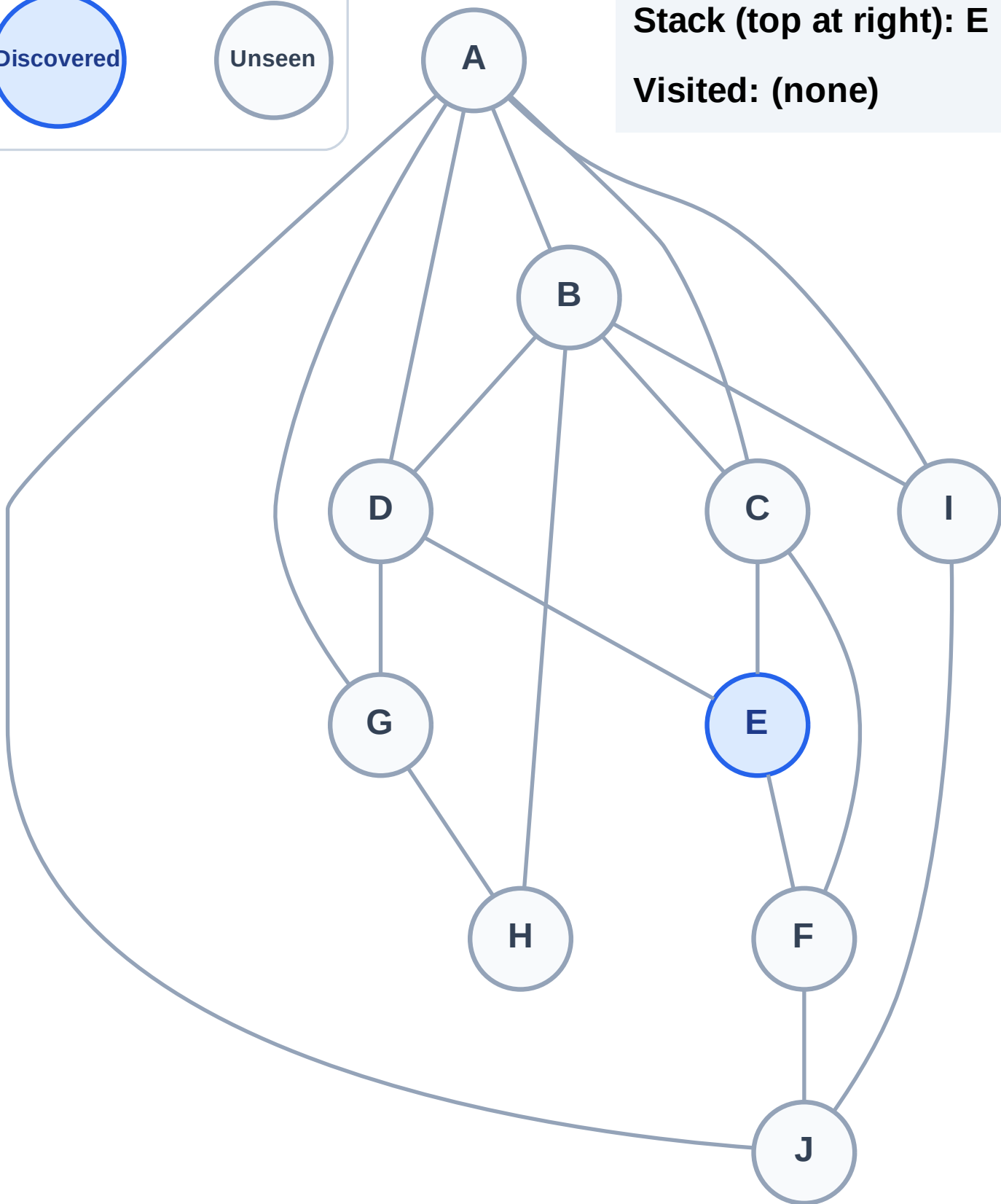
Step 1: Start at E

## Legend



Stack (top at right): E

Visited: (none)



# DFS Traversal

Step 2: Process node E

## Legend

Complete

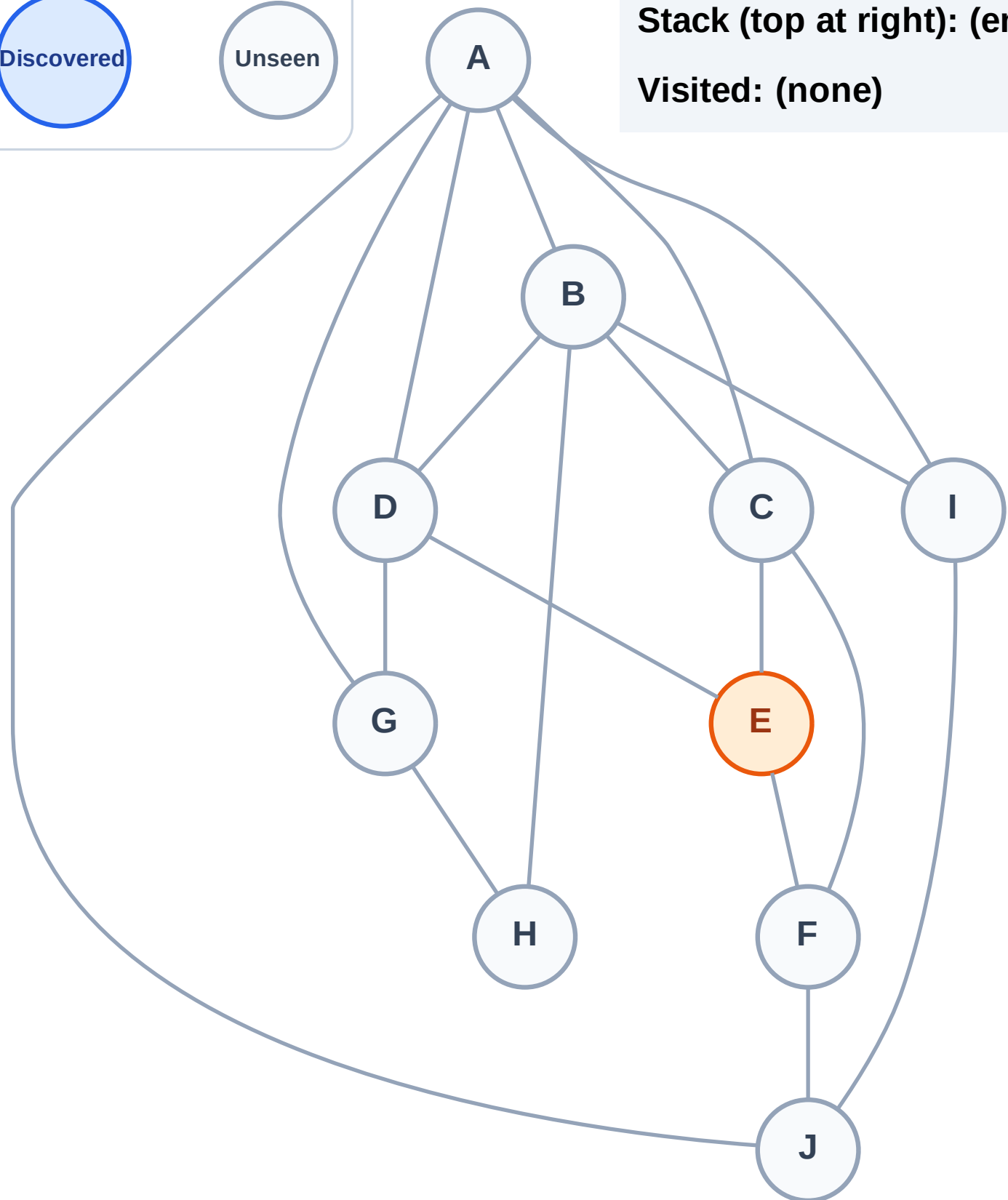
Processing

Discovered

Unseen

Stack (top at right): (empty)

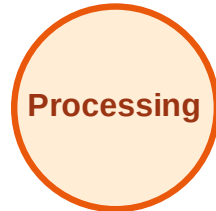
Visited: (none)



# DFS Traversal

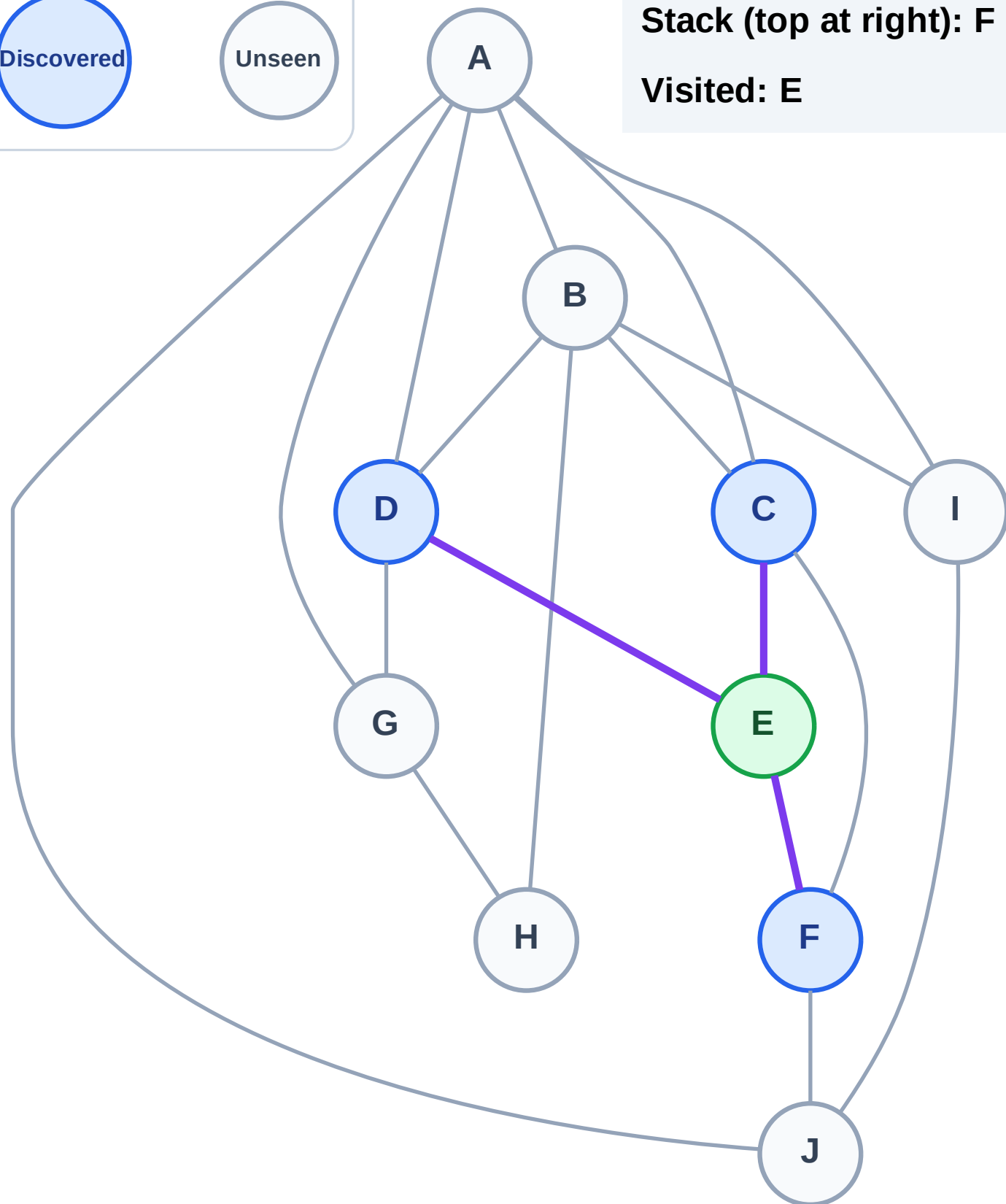
Step 3: Discover F, D, C from E

## Legend



Stack (top at right): F | D | C

Visited: E



# DFS Traversal

Step 4: Process node C

Legend

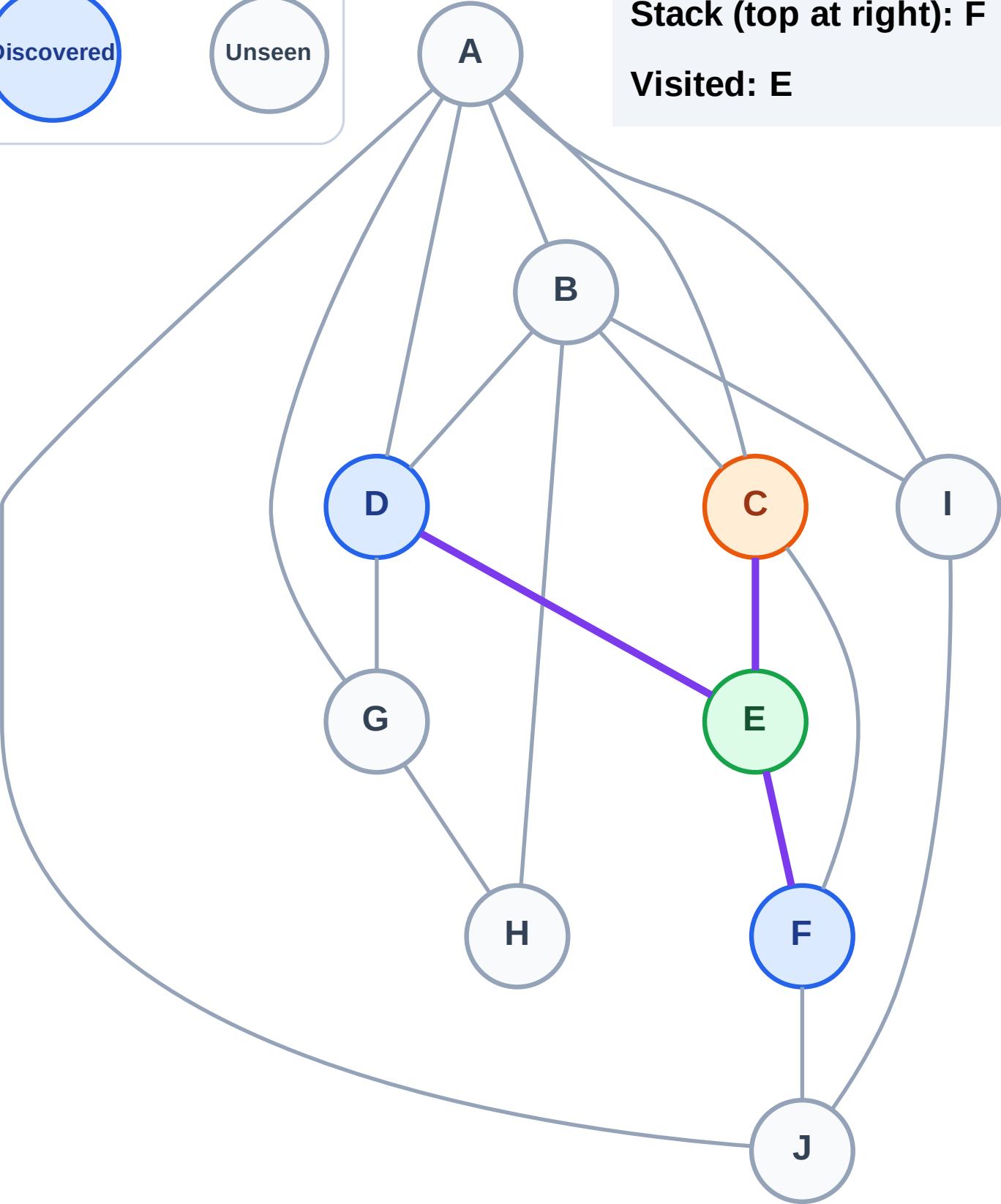
Complete

Processing

Discovered

Unseen

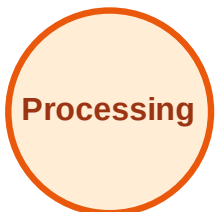
Stack (top at right): F | D  
Visited: E



# DFS Traversal

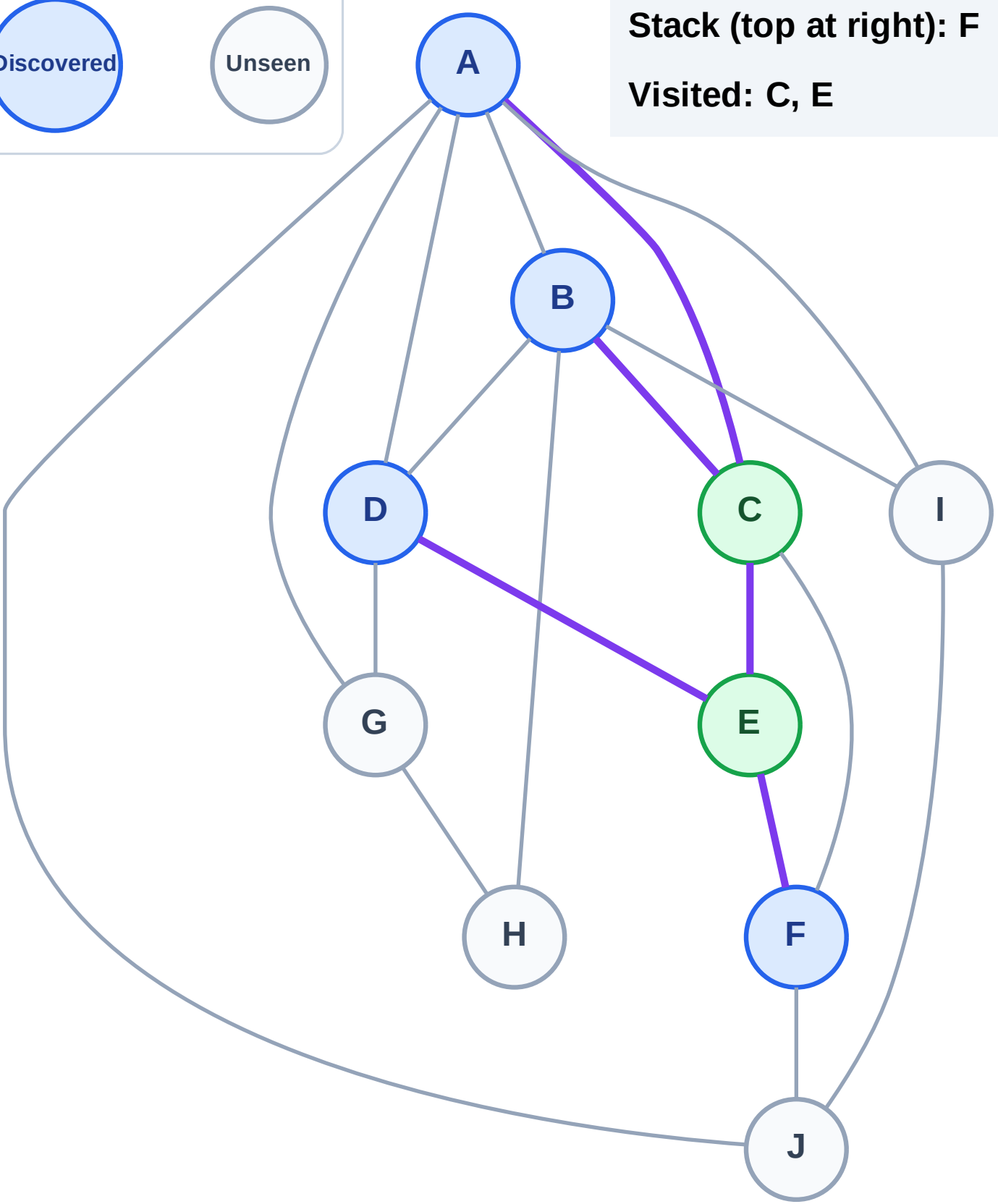
Step 5: Discover B, A from C

## Legend



Stack (top at right): F | D | B | A

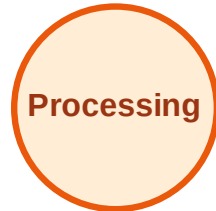
Visited: C, E



# DFS Traversal

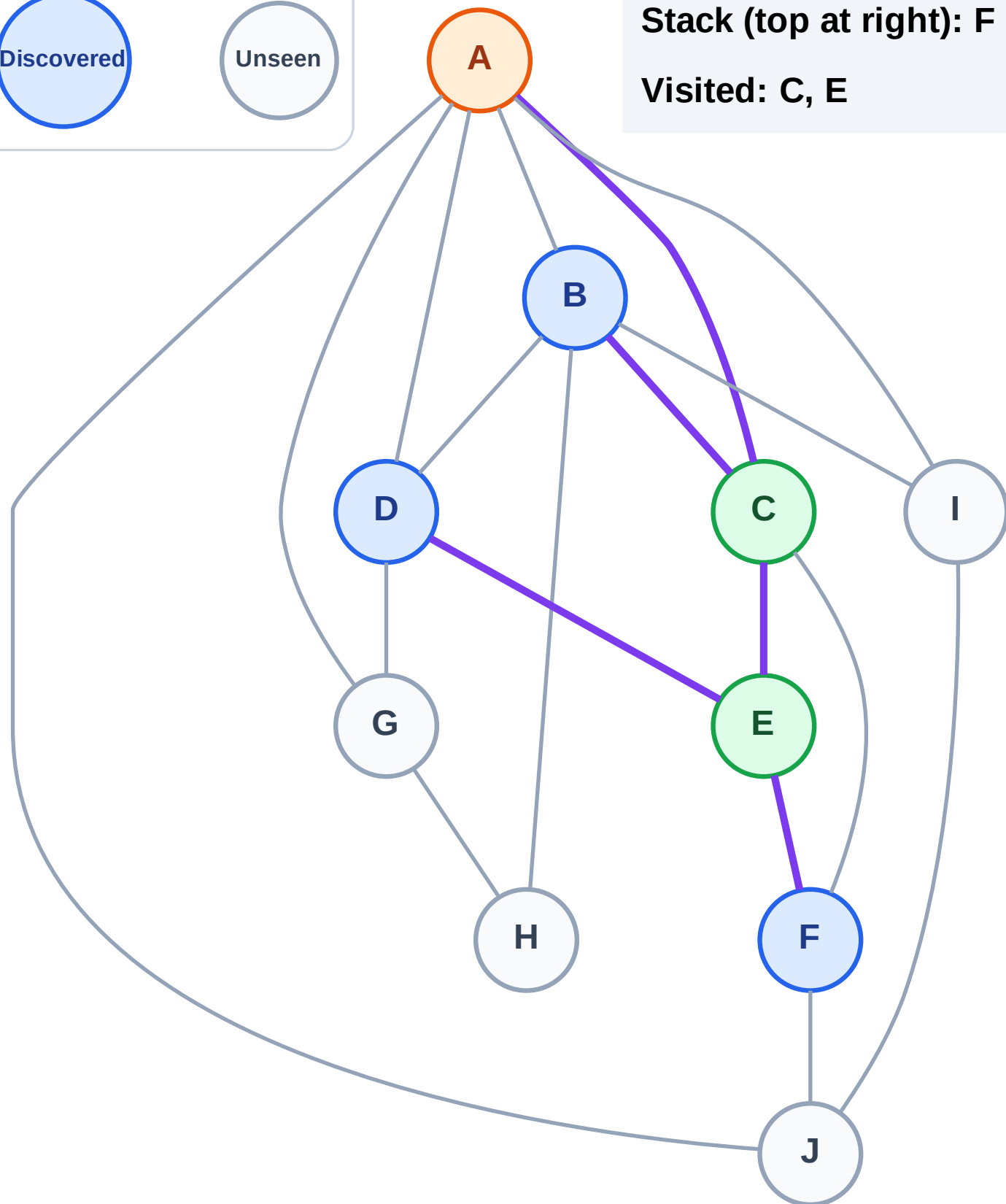
Step 6: Process node A

## Legend



Stack (top at right): F | D | B

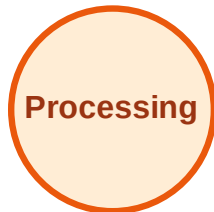
Visited: C, E



# DFS Traversal

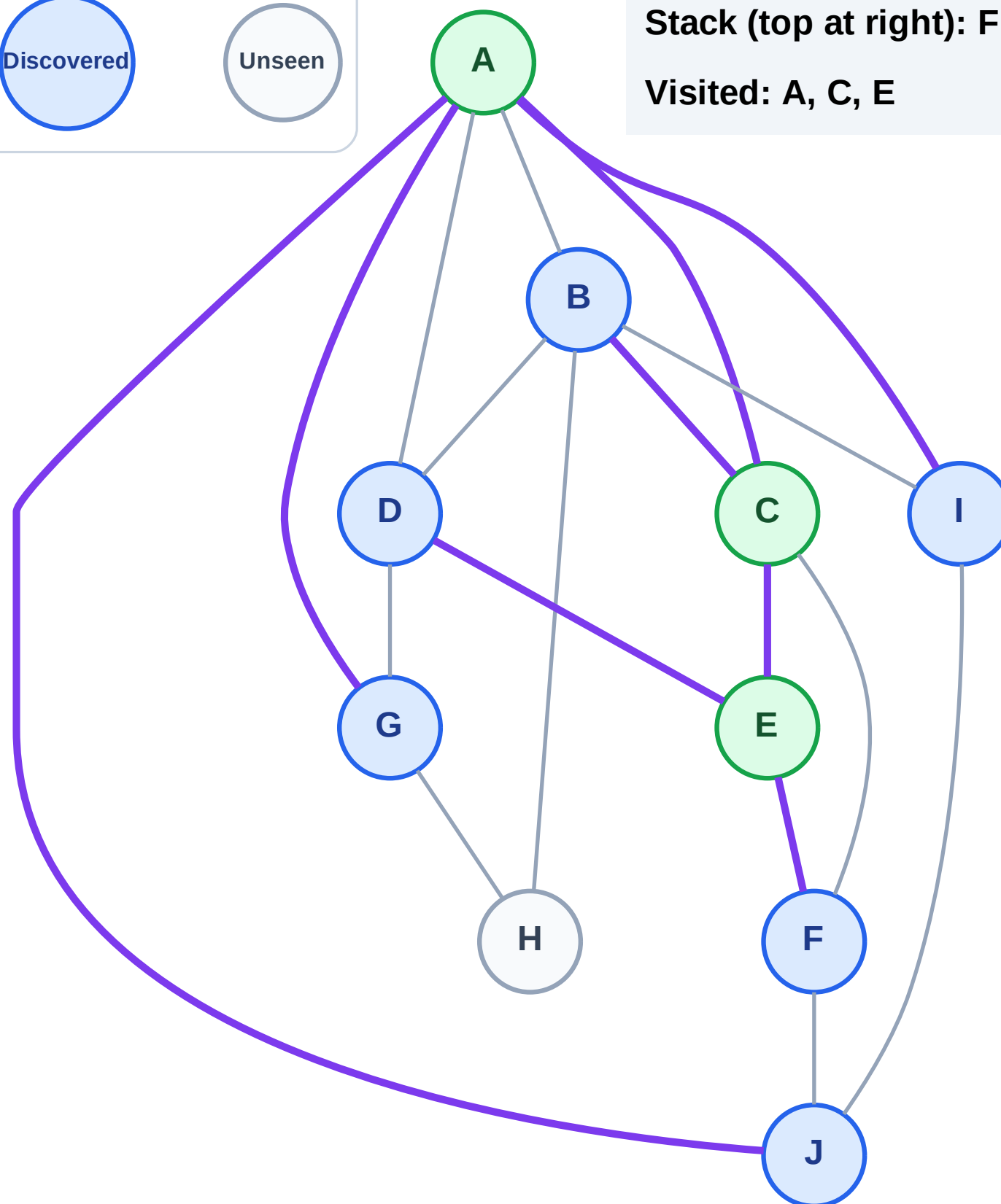
Step 7: Discover J, I, G from A

## Legend



Stack (top at right): F | D | B | J | I | G

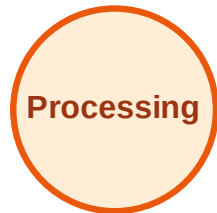
Visited: A, C, E



# DFS Traversal

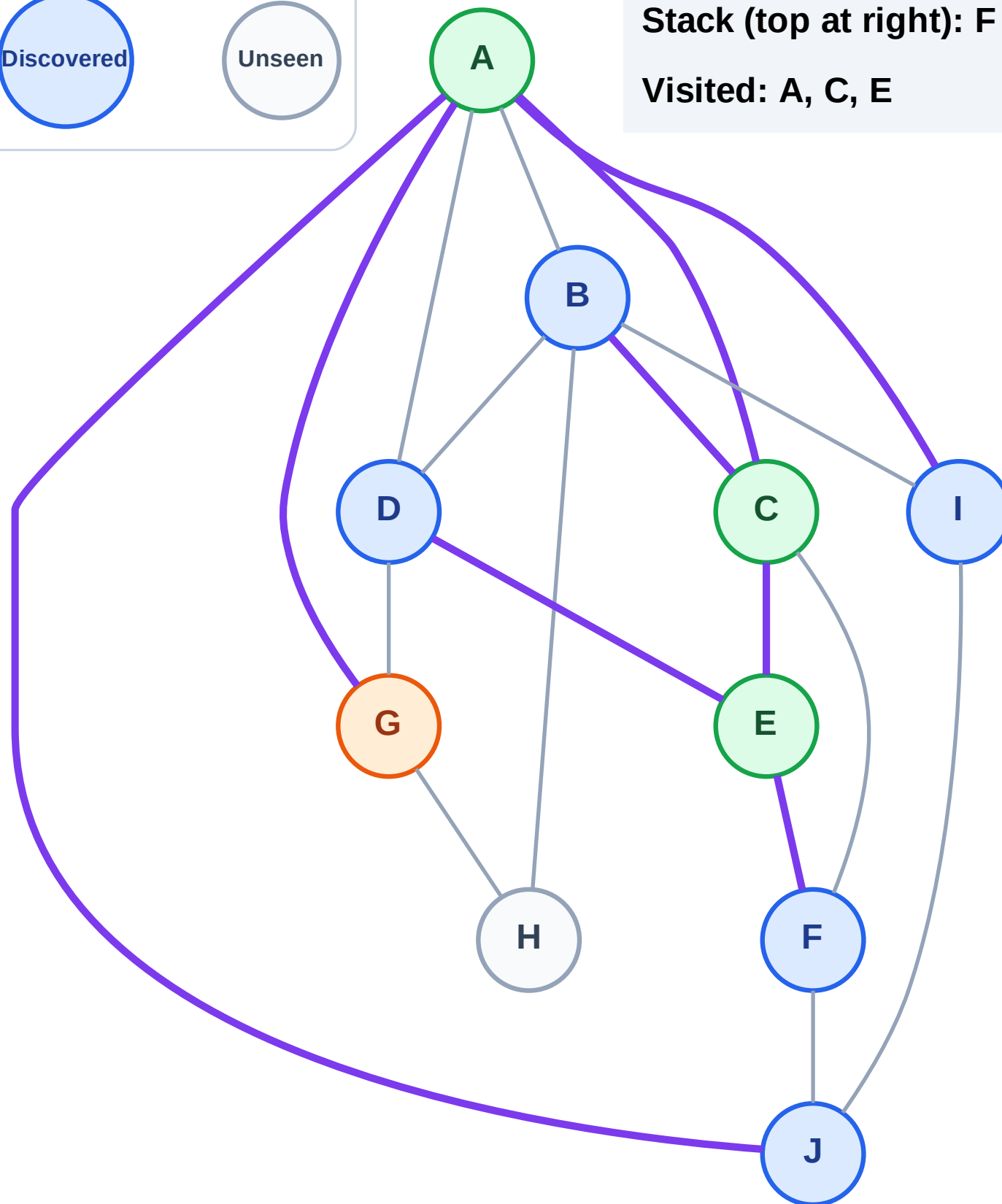
Step 8: Process node G

## Legend



Stack (top at right): F | D | B | J | I

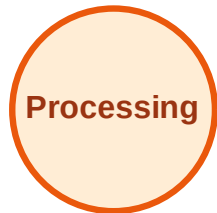
Visited: A, C, E



# DFS Traversal

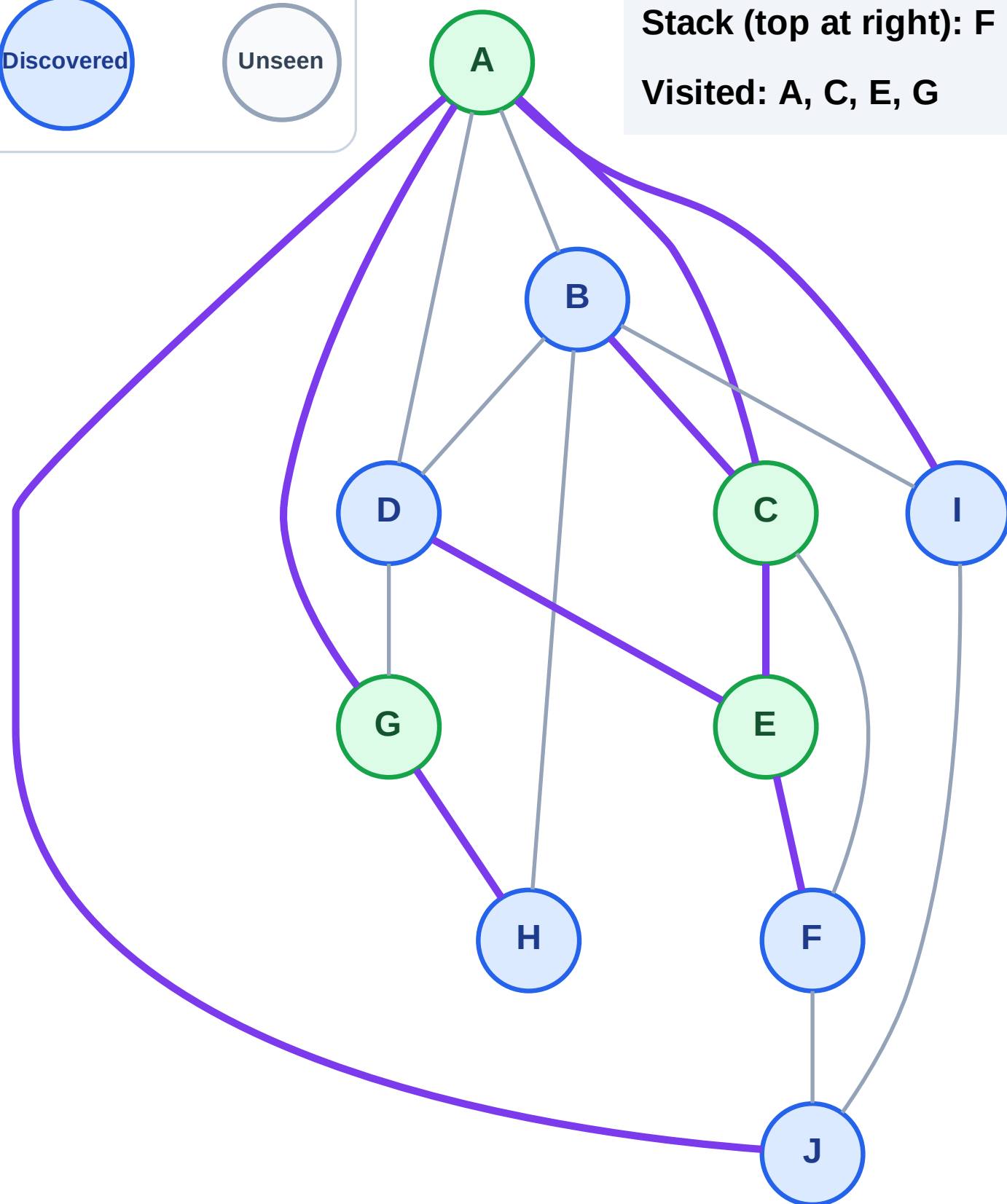
Step 9: Discover H from G

## Legend



Stack (top at right): F | D | B | J | I | H

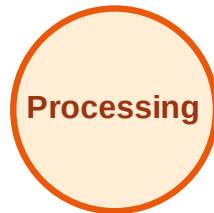
Visited: A, C, E, G



# DFS Traversal

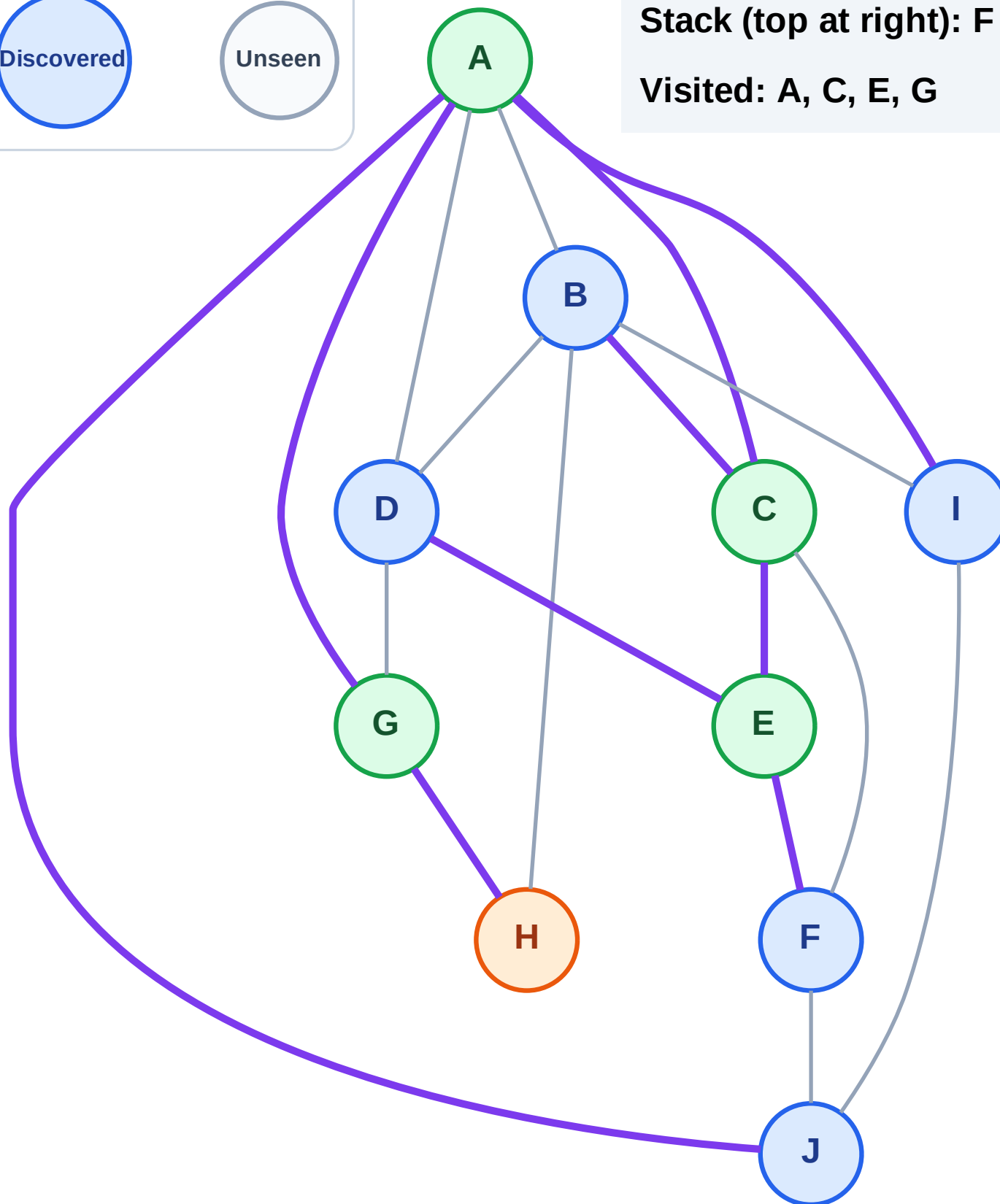
Step 10: Process node H

## Legend



Stack (top at right): F | D | B | J | I

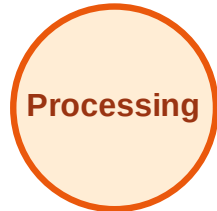
Visited: A, C, E, G



# DFS Traversal

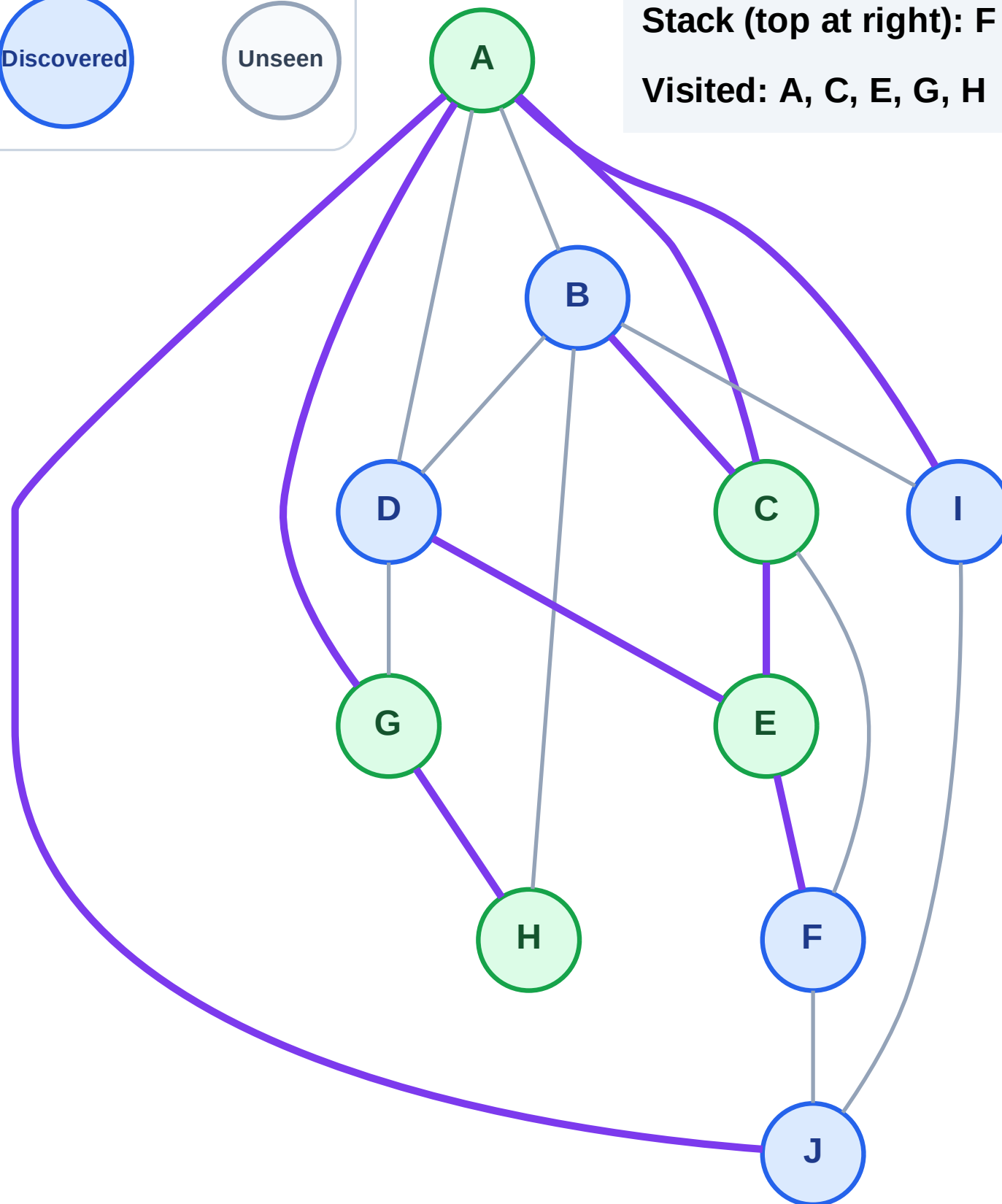
Step 11: No new neighbors from H

## Legend



Stack (top at right): F | D | B | J | I

Visited: A, C, E, G, H



# DFS Traversal

Step 12: Process node I

Legend

Complete

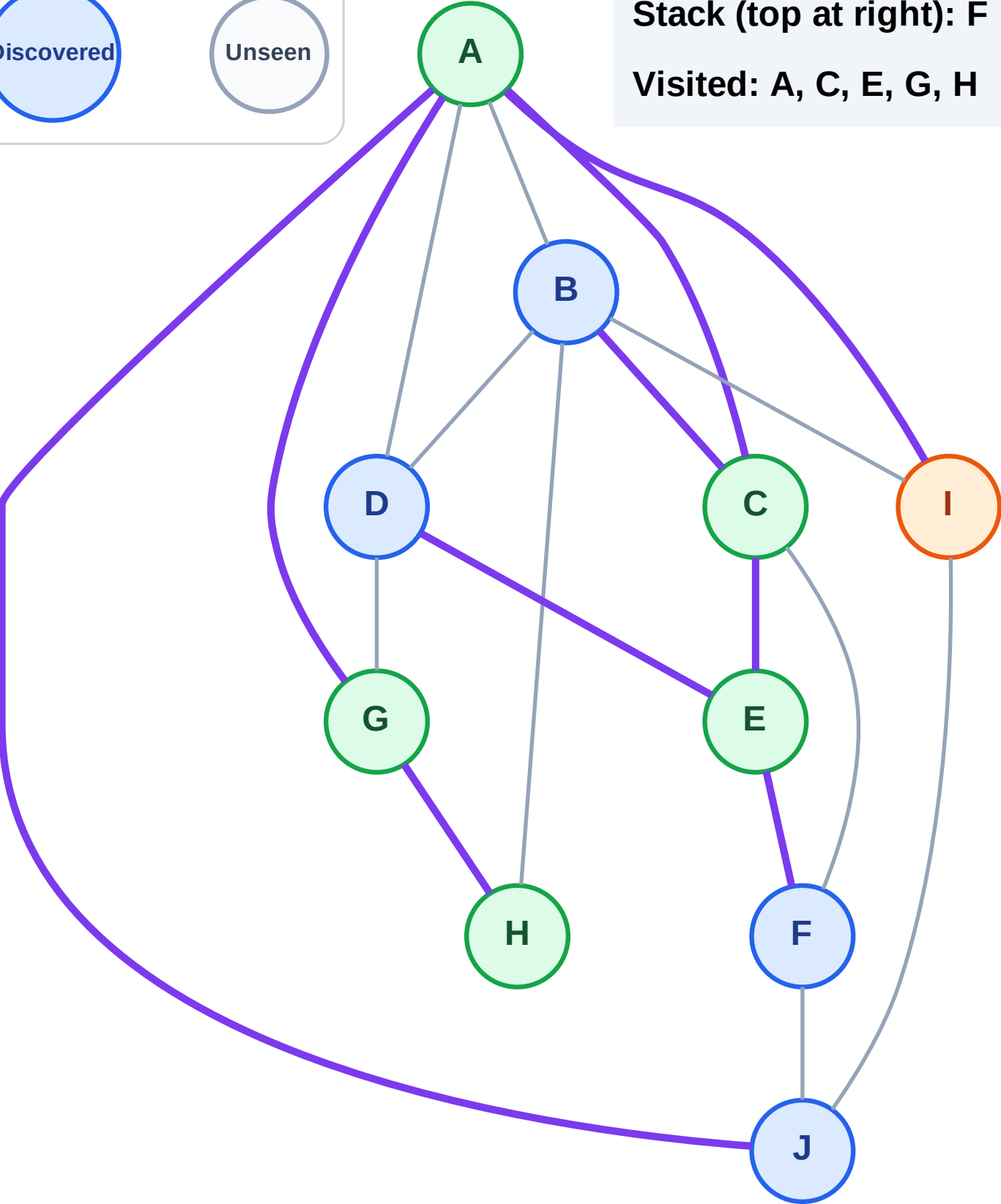
Processing

Discovered

Unseen

Stack (top at right): F | D | B | J

Visited: A, C, E, G, H



# DFS Traversal

Step 13: No new neighbors from I

Legend

Complete

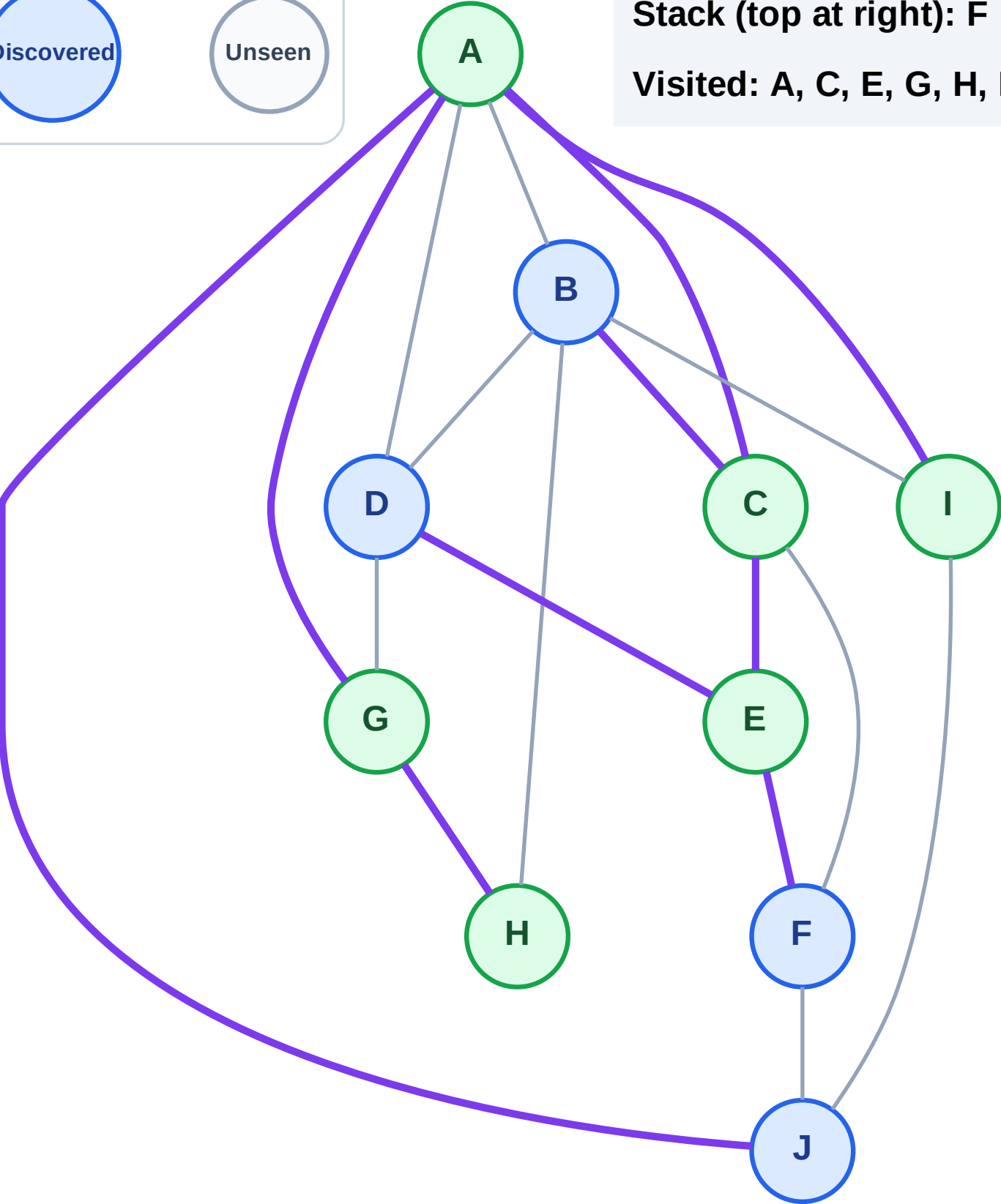
Processing

Discovered

Unseen

Stack (top at right): F | D | B | J

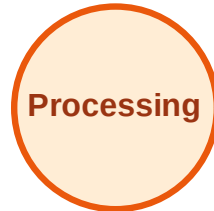
Visited: A, C, E, G, H, I



# DFS Traversal

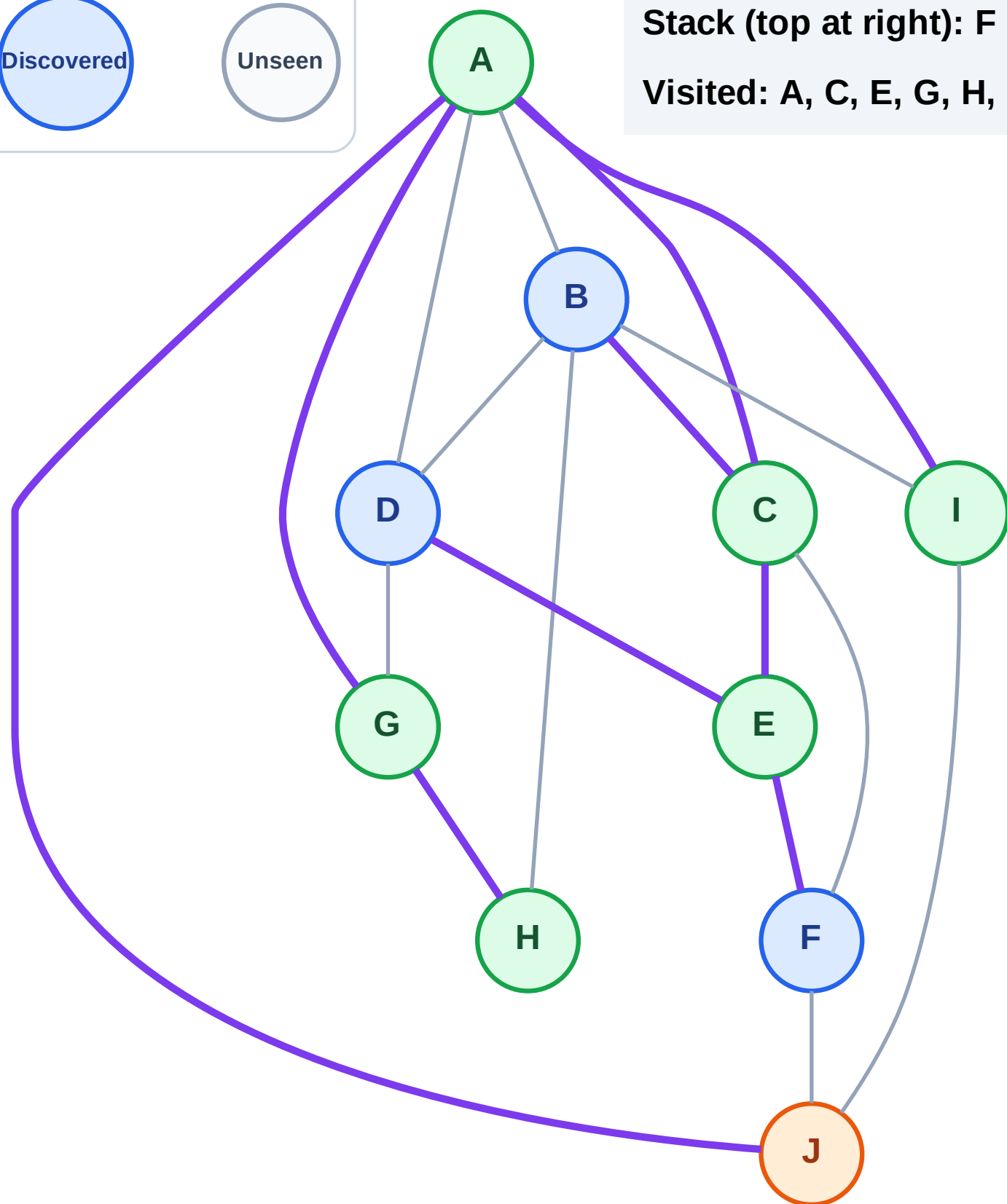
Step 14: Process node J

## Legend



Stack (top at right): F | D | B

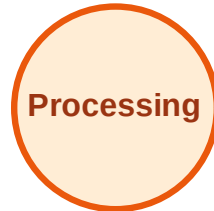
Visited: A, C, E, G, H, I



# DFS Traversal

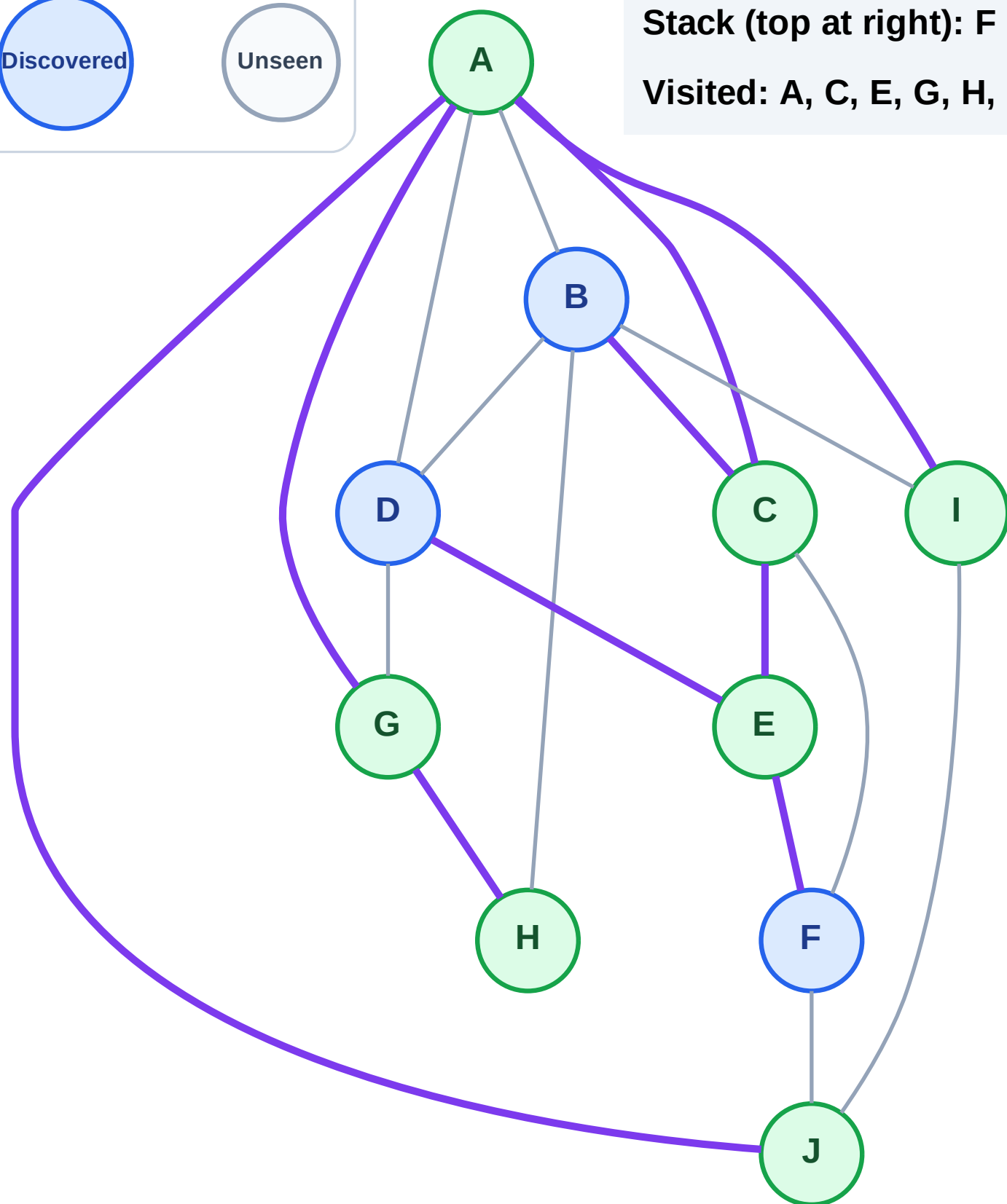
Step 15: No new neighbors from J

## Legend



Stack (top at right): F | D | B

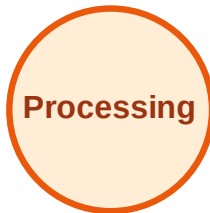
Visited: A, C, E, G, H, I, J



# DFS Traversal

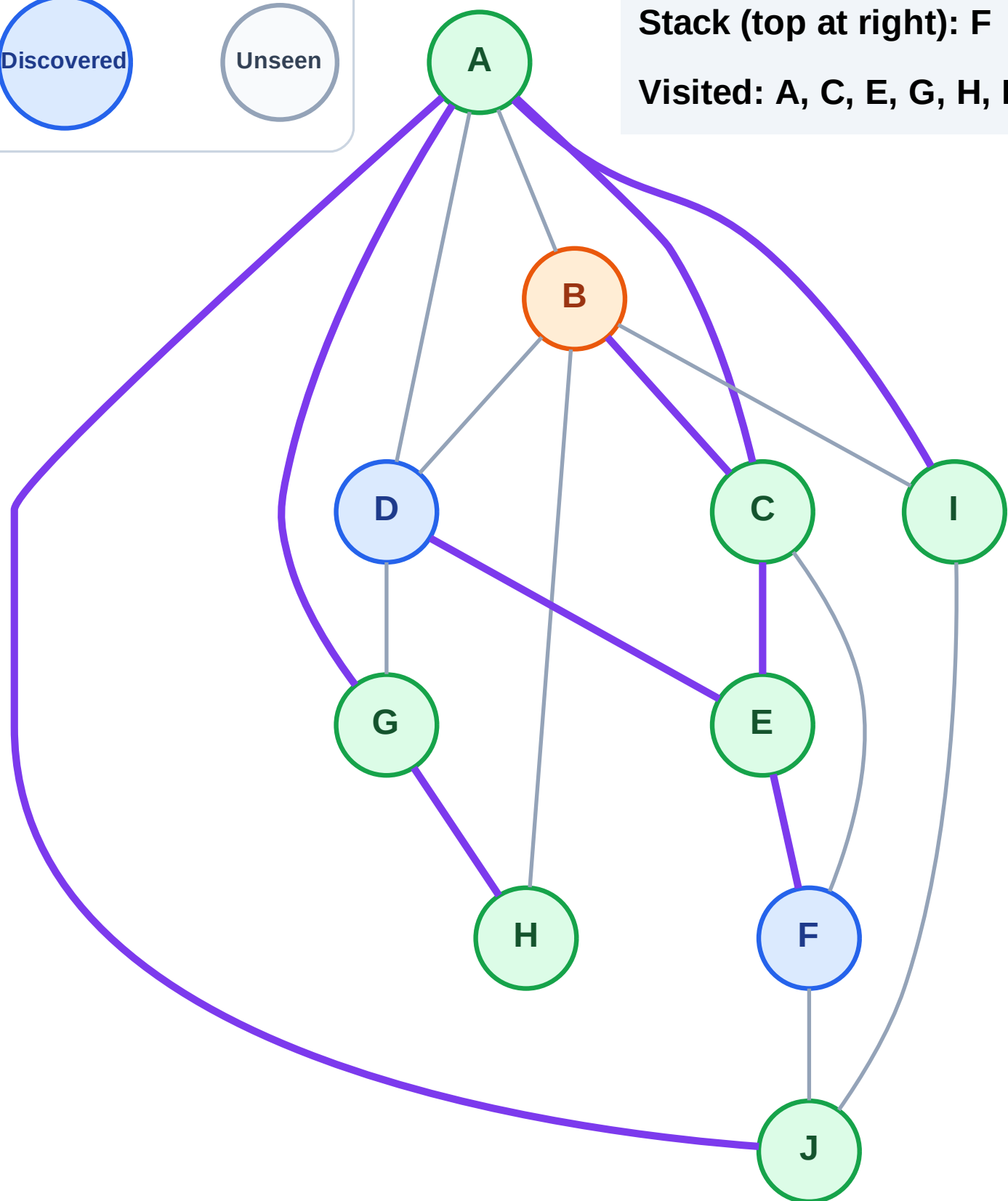
Step 16: Process node B

## Legend



Stack (top at right): F | D

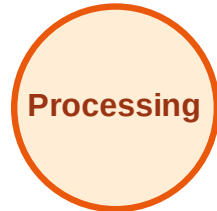
Visited: A, C, E, G, H, I, J



# DFS Traversal

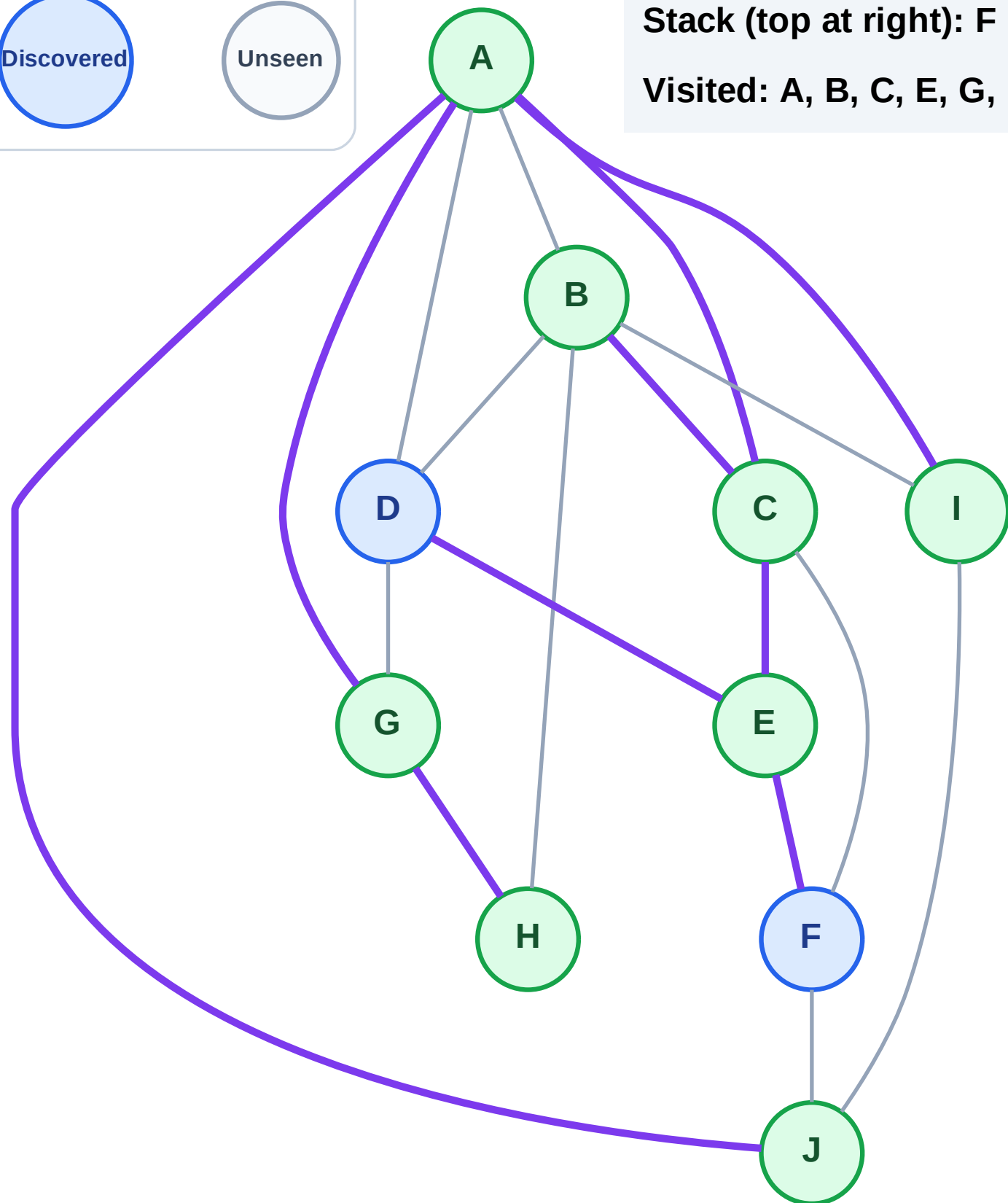
Step 17: No new neighbors from B

## Legend



Stack (top at right): F | D

Visited: A, B, C, E, G, H, I, J



# DFS Traversal

Step 18: Process node D

Legend

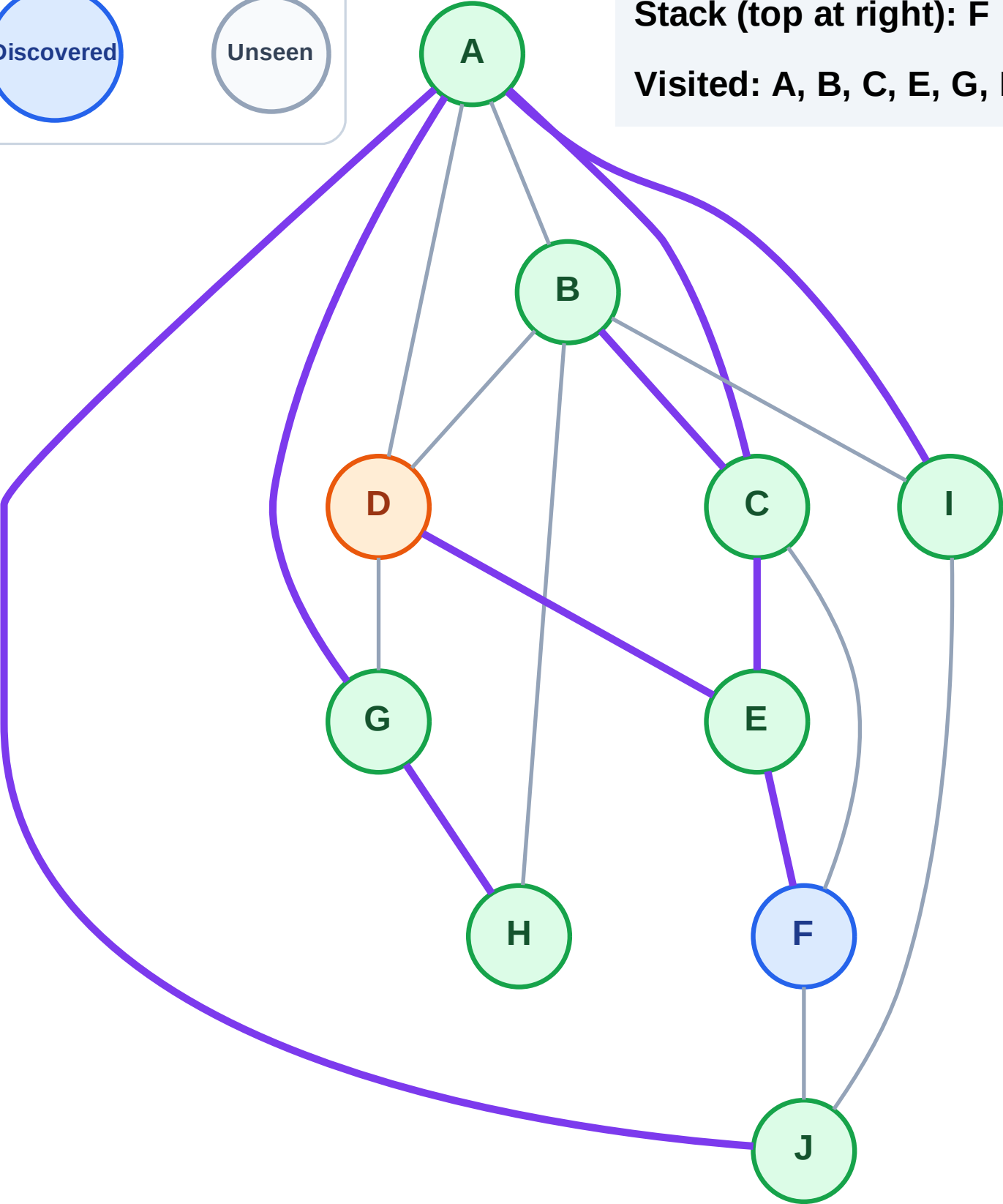
Complete

Processing

Discovered

Unseen

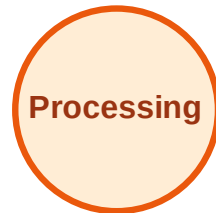
Stack (top at right): F  
Visited: A, B, C, E, G, H, I, J



# DFS Traversal

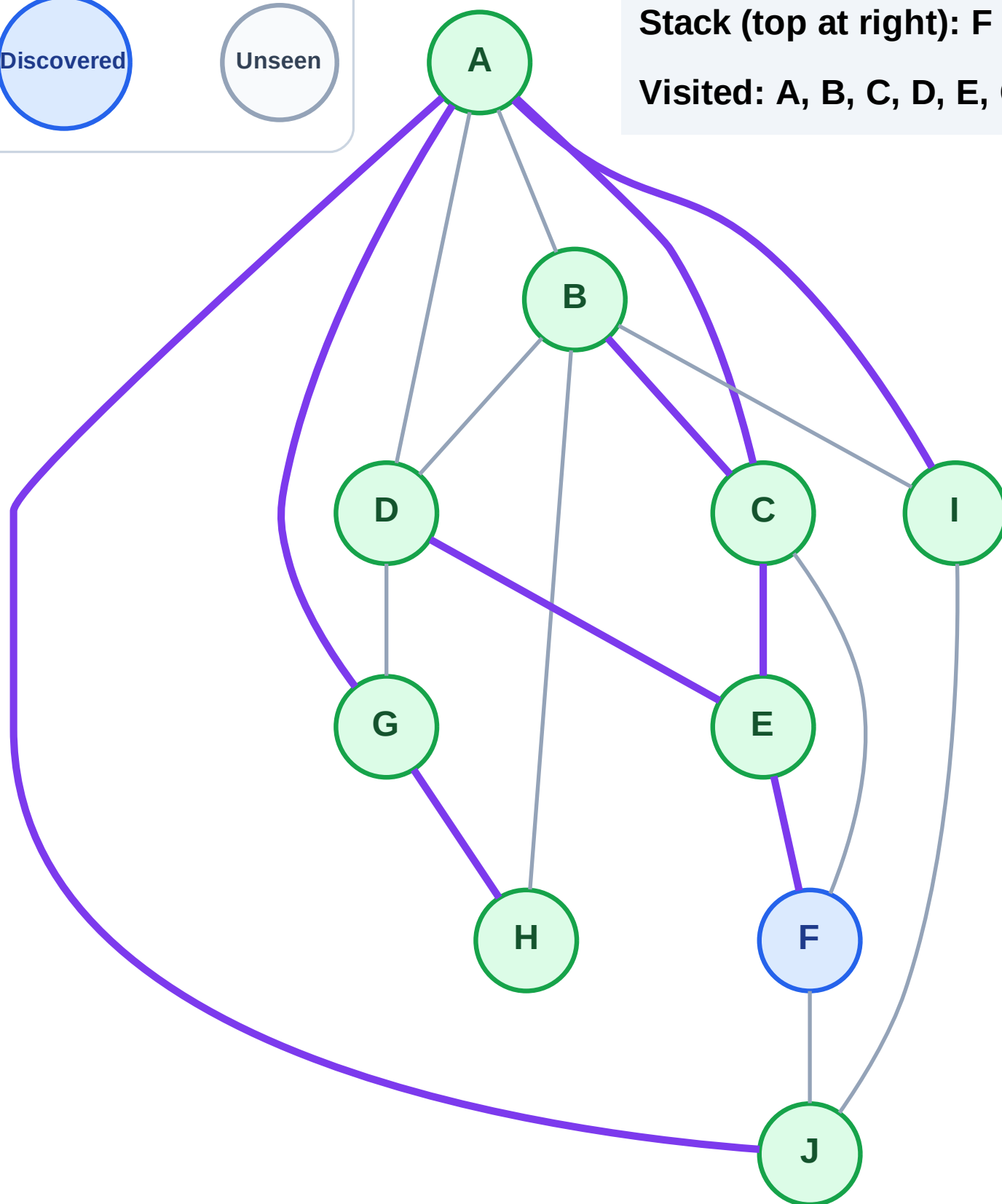
Step 19: No new neighbors from D

## Legend



Stack (top at right): F

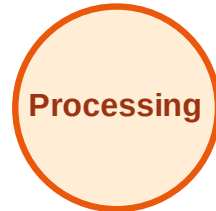
Visited: A, B, C, D, E, G, H, I, J



# DFS Traversal

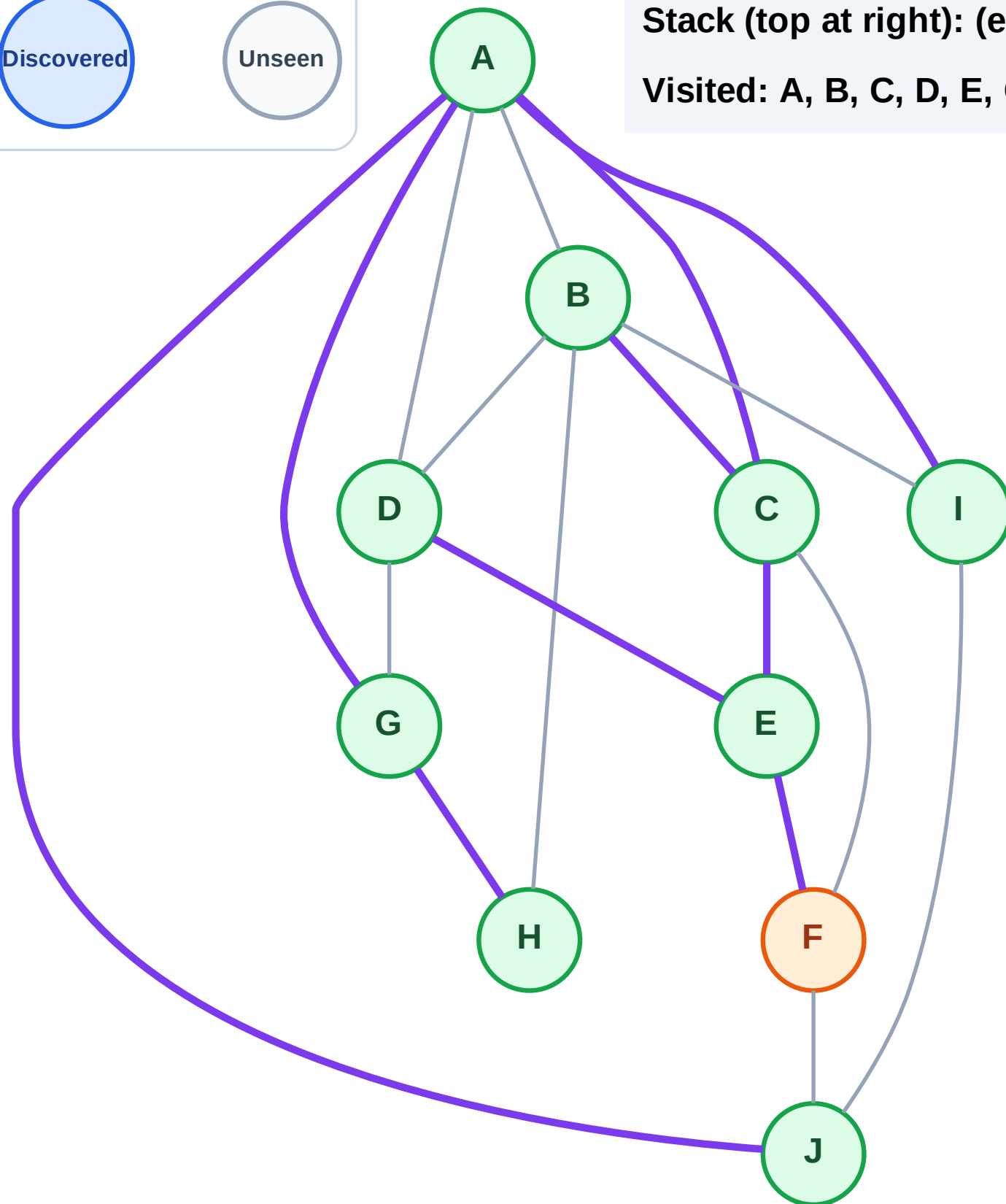
Step 20: Process node F

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, G, H, I, J



# DFS Traversal

Step 21: No new neighbors from F

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

